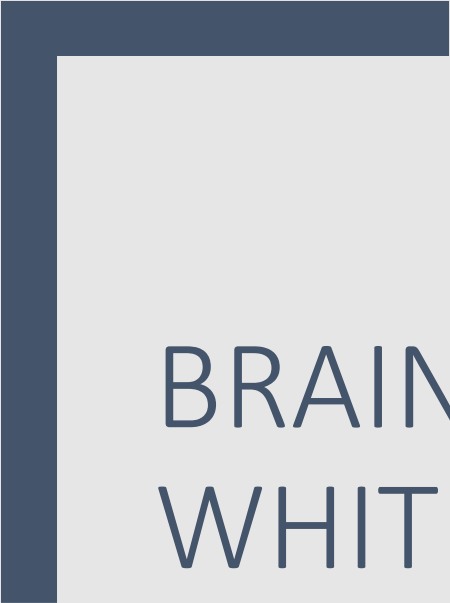




BRAINCOIN- WHITEPAPER

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Chapter 1 | Concept and Vision

第一章 | 概念与愿景

1.1 The Evolution of Human Labor | 人类劳动的演化

Throughout human history, the definition of labor has evolved in parallel with the development of civilization.

In the **industrial era**, labor was defined by **physical exertion**—the transformation of energy and matter through human strength and mechanical tools.

In the **information era**, labor became a function of **symbolic manipulation**—the management of data, communication, and computation.

Now, as artificial intelligence and automation replace both manual and informational work, a new form of labor emerges: **cognitive labor**, the act of thinking itself.

Cognitive labor is not a metaphor; it is a measurable phenomenon.

Neuroscience has demonstrated that mental activity manifests as quantifiable energy flow and neural network entropy within the brain.

What was once invisible—thought, imagination, insight—can now be observed and analyzed. Thus, for the first time in history, **the human mind becomes a productive entity** within the measurable economy.

在人类文明的发展进程中，“劳动”的定义始终与时代的生产方式相伴演化。

在工业文明中，劳动是**体力的转化**——人类通过机械与能源改变物质世界；

在信息文明中，劳动是**符号的操控**——人类通过数据与知识创造价值。

而当人工智能与自动化逐步取代了体力与信息劳动后，

一种新的劳动形态开始浮现：**认知劳动（Cognitive Labor）**，即思考本身的劳动。

认知劳动并非抽象隐喻，而是一种可被观测与计量的生理与信息现象。

神经科学揭示：思维的过程伴随脑电能流、突触放电频率与神经网络熵值的变化。

曾经不可见的“思考”、“灵感”与“洞察”，如今已具备科学可测性。

这意味着：**人类的大脑，首次成为可被量化的生产主体。**

1.2 The End of Traditional Currency | 传统货币的终局

Traditional currencies—gold, paper, or digital—are all rooted in the logic of **scarcity**.

They derive value from control over limited physical or symbolic resources.

However, in a world where material scarcity is being replaced by cognitive abundance, the logic of money reaches its epistemological limit.

When machines can autonomously produce goods, and algorithms can autonomously generate knowledge, the only remaining scarce resource is **conscious attention**. This is the ultimate frontier of economic measurement.

传统意义上的货币——无论是黄金、纸币，还是加密货币——其价值均建立在“**稀缺性（Scarcity）**”之上。货币体系通过控制有限的物质或符号资源来定义交换与财富。然而，当物质稀缺被技术所克服、当信息生产由算法自动完成，经济体系的核心稀缺性已然转移——唯一真正稀缺的资源，是**意识（Consciousness）**本身。

这标志着传统货币逻辑的终结：
货币的未来，不再取决于物质的拥有，而取决于思维的存在。

1.3 Thought as Labor | 思考即劳动

The central premise of BrainCoin is that **thinking constitutes labor**. Every act of cognition—whether creative, analytical, or contemplative—represents measurable effort, and therefore, legitimate value generation.

By integrating brain-computer interfaces (BCI) with blockchain infrastructure, BrainCoin introduces a **Proof of Brain Work (PoBW)** mechanism: neural activity becomes the measurable proof of mental contribution. This establishes a new category of economy, where **conscious energy replaces mechanical energy** as the foundation of productivity.

脑币体系的核心假设是：**思考即劳动，思维即价值**。
每一次认知活动——无论是创造、分析还是反思——都代表一种可被计量的脑能量消耗，因此具备劳动与报酬的合理性。

通过将脑机接口（BCI）与区块链系统结合，脑币提出了“**脑力工作量证明（Proof of Brain Work, PoBW）**”机制：以脑电与神经信号为依据，对思维活动进行验证与记录。这使得“生产力”的概念第一次从物理能量转向**意识能量（Conscious Energy）**。

1.4 From Energy Economy to Cognitive Economy | 从能量经济到认知经济

The industrial economy transformed matter.
The digital economy transformed information.
The cognitive economy transforms **meaning**.

In this new paradigm, productivity is no longer determined by time, machinery, or data throughput,
but by the **authenticity and complexity of mental activity**.
BrainCoin positions itself as the fundamental measurement unit of this cognitive economy—a currency of thought, not transaction.

工业经济改变了物质，
信息经济改变了数据，
而**认知经济**将改变意义本身。

在这种新范式下，生产力的决定因素不再是时间、机器或信息量，
而是**思维活动的真实性与复杂度**。
脑币由此成为认知经济的基本计量单位——
一种不以交易为目的的货币，而是以**思维本身为价值核心**的货币。

1.5 The Measurability of Consciousness | 意识的可计量性

Modern neurotechnology enables the non-invasive measurement of brain activity through EEG, fNIRS, and neural entropy analysis.
These advances allow the transformation of abstract cognitive effort into quantitative metrics.
However, this transition demands a new ethical paradigm:
measurement without surveillance, validation without exposure.

现代神经技术的发展（EEG、fNIRS、神经熵分析）
使得意识活动可以在非侵入式条件下被精确量化。
抽象的认知劳动因此具备了可计量性。
然而，这种计量必须受到严格伦理约束：
测量而不监控，验证而不暴露。

1.6 Core Principles of BrainCoin | 脑币的核心原则

BrainCoin rests upon four ethical and structural principles:

Principle	Description
Decentralized Cognition	Thought is distributed; no entity owns consciousness.
Measurable Authenticity	Only genuine neural activity generates value.
Non-transferable Equity	Wealth cannot be accumulated, only created and consumed.
Neuro-ethical Governance	Data sovereignty and cognitive health are inviolable.

脑币体系建立在以下四项原则之上：

原则	含义
去中心化意识	思维是分布式存在，无人可垄断意识。
真实可测性	仅真实的神经活动才具有价值。
不可转移平权	财富不再积累，而在创造与消耗中循环。
神经伦理治理	意识隐私与心理健康不可侵犯。

1.7 Vision: Toward a Conscious Civilization | 愿景：走向意识文明

BrainCoin envisions a global system in which economic value aligns with cognitive integrity, and where humanity evolves from an economy of consumption to an ecology of consciousness.

In this civilization,
thinking becomes the universal form of contribution,
awareness becomes the measure of equality,
and consciousness becomes the foundation of law.

脑币的愿景是建立一个**价值与意识相一致**的全球体系，
让人类从“消费经济”迈向“意识生态”。

在这种文明形态中：
思考成为普遍的贡献方式，
意识成为平等的衡量标准，
而认知成为法律与社会秩序的根基。

Summary Matrix | 总结矩阵

Concept 概念	Transition 转变
Labor 劳动	From physical exertion to cognitive activity 从体力付出到认知活动

Concept 概念	Transition 转变
Value 价值	From scarcity of resources to scarcity of consciousness 从资源稀缺转向意识稀缺
Technology 技术	From automation to neuro-integration 从自动化转向神经一体化
Economy 经济	From transaction to cognition 从交易逻辑转向认知逻辑
Civilization 文明	From control to co-creation 从控制型社会迈向共创型文明

Selected Sources (academic) | 参考资料（学术为主）

These works ground the shifts from physical to cognitive labor, and from automation to neuro-measurement:

- Rifkin, J. (2014). *The Zero Marginal Cost Society*. Palgrave Macmillan — argues the rising centrality of creative/cognitive production.
- Nunez, P. L., & Srinivasan, R. (2021). *Electric Fields of the Brain* (3rd ed.). OUP — neurophysics foundations of EEG measurement.
- Cohen, M. X. (2014). *Analyzing Neural Time Series Data*. MIT Press — methods for EEG power/entropy metrics used to quantify cognitive states.

下列文献为从体力到认知劳动、从自动化到神经计量的转变提供学术支撑：

- Rifkin, J. (2014) 《零边际成本社会》— 提出现代经济中创造性/认知生产的重要性上升。
- Nunez & Srinivasan (2021) 《大脑的电场（第三版）》— EEG 神经物理学基础。
- Cohen (2014) 《神经时间序列分析》— 用于量化认知状态的 EEG 功率/熵等方法框架。

Chapter 2 | Token Generation Mechanism & Technical Feasibility

第二章 | 发币机制与技术可行性

EN — What changed in this revision: Adds rigorous measurement rules (TMV-Spec), three-factor authenticity, baseline normalization, artifact penalties, BRC-coupled reward caps, hash-only on-chain commitments, ZK-Receipts, quantitative KPIs for feasibility, and a “Minimal Viable PoNP” profile.

****中 — 本版改动要点：****新增 TMV-Spec 计量规范、三因子真实性门、个体基线归一化、伪迹惩罚、与 BRC 耦合的奖励上限、仅哈希上链与零知识收据、可量化 KPI 及“最小可行 PoNP”配置。

2.1 Conceptual Foundation: Proof of Brain Work (PoBW)

2.1 概念基础：脑力工作量证明（PoBW）

EN — BrainCoin introduces Proof of Brain Work (PoBW): a consensus mechanism that converts **measurable** neural activity into digital currency. Unlike Bitcoin’s **Proof of Work (PoW)**, which prices **computational energy**, PoBW prices **cognitive energy** using non-invasive EEG-based brain-computer interfaces (BCIs). This is a shift **from energy-based mining to mind-based mining**, where **human thought** is the unit of economic value.

中 — 脑币引入脑力工作量证明（PoBW）：将可测量的神经活动转换为数字货币的共识机制。不同于比特币以**计算能量**为核心的 **PoW**，PoBW 以非侵入式 EEG 脑机接口**计量认知能量**。这标志着从**能量挖矿**到**心智挖矿**的转变，**人类思维**成为价值计量单位。

2.2 Technical Architecture: From Neural Signals to Tokens

2.2 技术架构：从脑信号到代币

System Overview

系统架构概览

EN — EEG Headset → Signal Processing → Cognitive Load Model → Authenticity Gate → Blockchain Proof → Token Reward

中 — EEG 头戴设备 → 信号处理 → 认知负荷模型 → 真实性门 → 区块链证明 → 代币奖励

EEG / BCI Layer

脑机接口层 (EEG / BCI)

EN — Use **non-invasive EEG** headsets to acquire brain signals. Recommended: sampling ≥ 256 Hz, **8–16** channels minimum (10–20 montage incl. Fz/Cz/Pz/Oz). Canonical bands: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), γ (>30 Hz).

中 — 使用非侵入式 EEG 设备采集脑信号。建议采样率 ≥ 256 Hz，最少 **8–16** 通道（10–20 体系含 Fz/Cz/Pz/Oz）。常用频段： δ (0.5–4 Hz)、 θ (4–8 Hz)、 α (8–13 Hz)、 β (13–30 Hz)、 γ (>30 Hz)。

Signal Processing Layer

信号处理层

EN — Preprocess with **0.5–45 Hz** band-pass and **50/60 Hz** notch; remove ocular/muscle artifacts using **ICA**; segment into **2–4 s** epochs (50% overlap). Compute features: **power spectral density (PSD)**, **entropy** (e.g., Shannon/LZ), **inter-regional coherence**, and optionally **spectral slope (1/f)**.

中 — 采用 **0.5–45 Hz** 带通与 **50/60 Hz** 陷波滤波；用 **ICA** 去除眼动/肌电伪迹；切分为 **2–4** 秒短窗（50% 重叠）。提取特征：功率谱密度（PSD）、熵（如 Shannon/LZ）、脑区相干性，及可选的谱斜率（1/f）。

TMV-Spec (Task–Metric–Validity)

TMV-Spec (任务–指标–效度)

EN — Each earning task must declare: (a) **task archetype** (focus/learning/creation/collaboration), (b) **primary metrics** (band powers P_i , entropy H , optional slope/LZ), (c) **validity evidence** (peer-reviewed references), (d) **reliability targets** ($ICC \geq 0.6$), and (e) **failure controls** (artifact quota, uncertainty gating).

中 — 每个产币任务需明确：(a) 任务类型（专注/学习/创作/协作），(b) 主指标（频段功率 P_i 、熵 H ，可选斜率/LZ），(c) 效度证据（同行评审文献），(d) 信度目标（ $ICC \geq 0.6$ ），(e) 失效控制（伪迹配额与不确定性门控）。

High-frequency Guardrail (β – γ) — EMG-robust pipeline & residual checks

高频护栏 (β – γ) —— EMG 鲁棒流程与余留校验

EN — To protect authenticity and reward fairness, **high-frequency features (β - γ ; 13–45 Hz, esp. $\gamma > 30$ Hz) are admissible only** under an **EMG-robust** pipeline and are **never decisive alone** for the authenticity gate. Craniofacial muscles (frontalis/temporalis) can dominate power above ~20–30 Hz; therefore:

1. **Pipeline requirement.** Apply **0.5–45 Hz** band-pass + mains notch; run **ICA** (ocular + myogenic components), **muscle-spike detection/suppression**, and (where available) **EMG reference channels** for regression; segment 2–4 s with 50% overlap.
2. **Post-hoc EMG residual checks (must pass).** Governance-set thresholds on: **(i) spectral-slope test** (1/f slope not consistent with EMG-like flattening), **(ii) topography test** (no fronto-temporal dominance pattern typical of cranial EMG), **(iii) artifact-correlation test** (low correlation with jaw-clench/temporal sensors or EMG proxies).
3. **Exclusion rule.** If any residual check **fails**, **exclude γ -band** from **authenticity decisions** (it may remain in workload modeling with **zero weight** in the **Neural+1** gate). Even when admitted, γ -band contributes **capped weight** ($< w_{\gamma, \max}$) to **NeuralConfidence**.
4. **Minimum disclosure.** Sessions using γ -band must **publish EMG-residual metrics** (slope/topography/correlation) in the private audit log and expose an aggregate flag on-chain (no raw signals).

Rationale. High-frequency scalp activity is **readily contaminated** by cranial EMG; foundational studies show extensive **>20 Hz** contamination and characteristic **fronto-temporal** patterns, hence strict guardrails are standard practice. [PubMed+2PubMed+2](#)

中 — 为保障真实性与奖励公平，高频特征（ β - γ ；13–45 Hz，尤以 $\gamma > 30$ Hz）仅在通过 **EMG 鲁棒流程** 时方可纳入，且不得单独决定真实性门。由于面/颞肌（额肌/颞肌）可在约 **20–30 Hz** 以上主导功率，规则如下：

- 1) **流程要求：** 使用 **0.5–45 Hz** 带通 + 电源陷波；执行 **ICA**（去除眼动与肌电成分）、肌电尖峰检测/抑制，并在可用时引入 **EMG 参考通道回归**；2–4 秒窗、50% 重叠。
- 2) **后验 EMG 余留校验（必须通过）：** 治理设定阈值，包含：**(i) 频谱斜率**（不得呈现 EMG 式 1/f 扁平化）、**(ii) 脑地形检验**（不得出现 EMG 典型的额颞优势分布）、**(iii) 伪迹相关检验**（与咬肌/颞部传感或 EMG 代理的相关性需足够低）。
- 3) **剔除规则：** 任一余留校验未通过，则 γ 频段在真实性判定中不得使用（可保留在工作量建模，但在 **Neural+1** 门中的权重为零）。即便纳入， γ 的贡献也应封顶限权（ $< w_{\gamma, \max}$ ）。
- 4) **最小披露：** 使用 γ 特征的会话须在私有审计日志中记录 **EMG 余留指标**（斜率/拓扑/相关），并在链上公开聚合标记（不暴露原始信号）。

依据：高频头皮信号极易受颞肌 **EMG** 污染；经典研究显示 **>20 Hz** 普遍受扰，且呈额颞主导分布，故需严格护栏与报告。 [PubMed+2PubMed+2](#)

Cognitive Workload Model (PoBW core)

认知负荷模型（PoBW 核心）

EN — Define **Brain Work Index** W as a weighted combination of band powers and entropy with artifact penalty and personal normalization:

$$W = \sum_i \alpha_i P_i + \beta H, \tilde{W} = \frac{W - \mu_{\text{base}}}{\sigma_{\text{base}}}, W' = \tilde{W} - \lambda \cdot \text{ArtifactRate}.$$

Here P_i is power in band i ; H is entropy; $\mu_{\text{base}}, \sigma_{\text{base}}$ are **per-user baselines** (eyes-open rest, refreshed weekly); **ArtifactRate** is the fraction of rejected epochs; α_i, β, λ are learned/calibrated.
中 — 定义脑工作量指数 W 为频段功率与熵的加权，并加入个体归一化与伪迹惩罚：

$$W = \sum_i \alpha_i P_i + \beta H, \tilde{W} = \frac{W - \mu_{\text{base}}}{\sigma_{\text{base}}}, W' = \tilde{W} - \lambda \cdot \text{ArtifactRate}.$$

其中 P_i 为第 i 频段功率， H 为熵， $\mu_{\text{base}}, \sigma_{\text{base}}$ 为**个体基线**（每周更新的睁眼静息）；**ArtifactRate** 为伪迹窗比例； α_i, β, λ 为学习/校准参数。

Three-Factor Authenticity Gate (Liveness–Device–Task)

三因子真实性门（活体–设备–任务）

EN — Minting passes if Neural-plus-1 hold within a time-locked window:

(A) **Neural** — task-locked EEG markers (e.g., band power/spectral slope/LZ) meeting TMV thresholds;

(B) **Behavioral timing** — PVT/keystroke latencies with randomized jitter;

(C) **Device + watermark** — **TEE** attestation plus invisible stimulus watermarks (challenge–response) for replay/synthesis denial.

中 — 在时间锁定窗内满足神经+其一即通过：

(A) **神经学**——满足 TMV 阈值的任务耦合 EEG 指标（如频段功率/谱斜率/LZ）；

(B) **行为时序**——带随机抖动的 PVT/按键时延；

(C) **设备+水印**——**TEE** 认证与隐形刺激水印（挑战-响应）以拒绝重放/合成。

Blockchain Validation Layer (Commit-only + ZK)

区块链验证层（仅承诺上链 + 零知识）

EN — On-chain we **store only a hash** of the mint payload:

$$h = \text{Hash}(\text{epoch_nonce} \parallel \text{TEE_attest} \parallel \text{ZK_proof} \parallel R).$$

No raw EEG or sensitive metadata leave the device. Spending generates a **Zero-Knowledge Receipt (ZK-Receipt)** (one-time, non-transferable) proving **burn** $\geq$ **price** and optionally embedding an **SLA hash** for services.

中 — 上链仅提交哈希承诺：

$$h = \text{Hash}(\text{epoch_nonce} \parallel \text{TEE_attest} \parallel \text{ZK_proof} \parallel R)。$$

原始 EEG 与敏感元数据不出端。消费产生**零知识收据 (ZK-Receipt)**（一次性、不可转让），证明**销毁量** $\geq$ **价格**，并可嵌入 **SLA 哈希**。

Reward Layer (Health-capped log rule + NDA)

奖励层（健康封顶的对数规则 + 难度调节）

EN — Final reward uses a **logarithmic** schedule with **uncertainty gating**, **artifact penalty**, and **BRC cap**; **Neural Difficulty Adjustment (NDA)** stabilizes issuance:

$$\begin{aligned} R &= k_{\text{NDA}} \cdot \log \left[\frac{1}{2} (1 + \max(0, W')) \right] \cdot \mathbf{1}_{\text{pass}} \cdot \min(1, \text{BRCcap}), k_{\text{NDA}} \\ &= k \cdot f(\text{global median } W). \end{aligned}$$

$\mathbf{1}_{\text{pass}}$ requires authenticity gate + model confidence; **BRCcap** acts as a **limit**, not a multiplier boost.

中 — 最终奖励采用**对数**曲线，并施加**不确定性门控**、**伪迹惩罚**与 **BRC 上限**；以****神经网络难度调节 (NDA)** ******稳定总发放：

$$\begin{aligned} R &= k_{\text{NDA}} \cdot \log \left[\frac{1}{2} (1 + \max(0, W')) \right] \cdot \mathbf{1}_{\text{pass}} \cdot \min(1, \text{BRCcap}), k_{\text{NDA}} \\ &= k \cdot f(\text{global median } W)。 \end{aligned}$$

其中 $\mathbf{1}_{\text{pass}}$ 同时要求真实性门与模型置信；**BRCcap** 为**上限**而非**倍增器**。

2.3 Simplified Model: Proof of Neural Power (PoNP)

2.3 简化模型：神经功率证明（PoNP）

EN — Because current BCIs cannot decode **semantic content** reliably, BrainCoin supports a deployment-ready **PoNP** path that measures **how intensely** the brain is active, not **what it thinks**.
中 — 鉴于当前 BCI 难以稳定解码语义内容，脑币提供可立即部署的 **PoNP** 路径：度量思维强度而非思维内容。

EN — **Definition and computation.** Over a fixed window T , define **Brain Work Index (BWI)** as time-averaged EEG power; for sampled EEG $x[n]$ with N points:

$$\text{BWI} = \frac{1}{T} \int_0^T [V(t)]^2 dt \Leftrightarrow \text{BWI} = \frac{1}{N} \sum_{n=1}^N x[n]^2.$$

Personal normalization and artifact penalty follow Section 2.2: $\tilde{W}, W'$. Rewards use the same **health-capped log rule**.

中 — 定义与计算。在固定窗 T 内，以时间平均功率定义 **BWI**；离散形式为：

$$\text{BWI} = \frac{1}{T} \int_0^T [V(t)]^2 dt \Leftrightarrow \text{BWI} = \frac{1}{N} \sum_{n=1}^N x[n]^2.$$

并按 2.2 节进行个体归一化与伪迹惩罚（ $\tilde{W}, W'$ ），奖励遵循健康封顶对数规则。

EN — **Deployment profile.** Works on consumer-grade headsets (e.g., OpenBCI/Muse class) with standard filtering + ICA. No content decoding → **low privacy risk** and **fast rollout**.

中 — 部署概况。适配消费级头戴（如 OpenBCI/Muse 等），配合常规滤波 + ICA。因不解码内容 → 隐私风险低、落地快速。

EN — **Advantages.** Simple; hardware-feasible; robust to noise via time-averaging; compatible with authenticity gate when combined with behavioral timing and device watermarks.

中 — 优势。结构简单、硬件可行；时间平均具抗噪性；与行为时序与设备水印结合后可通过真实性门。

2.4 Technical Feasibility & Phase-Gate KPIs (2025)

2.4 技术可行性与阶段门槛（2025）

EN — **Replace star ratings with quantitative KPIs and acceptance gates.**

中 — 用可量化 KPI** 替代星级评价并设定准入门槛。 **

Layer	KPI (Acceptance Gate)	Key Limitation / Mitigation
EEG acquisition	SNR ≥ 6 dB; artifacted-epoch $\leq 15\%$; dropout $\leq 5\%$	Individual variability $\rightarrow$ personal baselines ; channel QC
Modeling (PoBW/PoNP)	Cross-subject AUC ≥ 0.75 (task vs. rest); ICC ≥ 0.6 test–retest	Limited EEG-only accuracy $\rightarrow$ optional slope/LZ; uncertainty gating
Authenticity (NVN)	Replay/synthesis TPR $\geq 95\%$ @ FPR $\leq 5\%$	Watermarks + TEE; challenge-response timing
Blockchain	Settlement P95 < 5 min (L2/batch); dispute re-mint < 24 h	Latency $\rightarrow$ rollups; offline queue with replay protection
Privacy	No raw EEG off device; DP-noised aggregates only	N/A

EN — **Phase-gate rule.** Advance to Phase-2 if $\geq 80\%$ of users across subgroups (gender/age/device type) meet **all KPIs** for two consecutive weeks.

中 — 阶段门槛。 若跨性别/年龄/设备等亚群的** $\geq 80\%$ 用户连续两周满足全部 KPI**， 则进入第二阶段。

EN — **Summary.** PoBW/PoNP are technically plausible now; industrial scale awaits better sensors and standards ($\approx 5\text{--}10$ years). Meanwhile BrainCoin functions as an **experimental parallel economy** with strong privacy and low energy footprint.

中 — 小结。 PoBW/PoNP 目前具原型可行性；大规模工业化仍待传感与标准进步（约 5–10 年）。过渡期内，脑币可作为并行实验经济运行，具较强隐私与低能耗特性。

2.5 Ethical & Systemic Reflections

2.5 伦理与系统反思

EN — **Effective labor.** Random thought and creative thinking are both **valuable**; **thought itself is labor**, echoing post-industrial work philosophy.

中 — 有效劳动。 随机思考与创造性思维皆为有价值的劳动，体现后工业劳动哲学。

EN — **Symbolism of energy.** Neural power is a **proxy for effort**, enabling **fairer distribution** that rewards **authentic cognitive engagement** over material throughput.

中 — 能量象征。神经功率作为努力的代理，以真实认知投入而非物质产出作为分配依据更为公平。

EN — Over-activation risk. Combine **hard hour caps** (≤ 8 h/day; ≤ 40 h/week) with **BRC** as a **limit scheduler** (not a multiplier).

中 — 过度激活风险。设定工时硬上限 (≤ 8 小时/日 ; ≤ 40 小时/周) , 并将 **BRC** 作为上限调度器 (非倍增器) 。

EN — Fairness & access. Publish inclusion metrics (median outputs by gender/age/disability/income); provide **device subsidies** and **public thinking hubs** to reduce access gaps.

中 — 公平与可及。公布包容性指标 (按性别/年龄/残障/收入的产出中位数) ; 提供设备补贴与公共思考空间以缩小门槛。

EN — Privacy by design. Local processing, DP-noised aggregates, **ZK-Receipts** for clearing; **hash-only** commitments on-chain.

中 — 隐私即设计。本地处理、差分隐私聚合、清算用 **ZK-Receipt** ; 上链采取仅哈希承诺。

2.6 Minimal Viable PoNP (Edge-Deployable Profile)

2.6 最小可行 PoNP (边缘可部署配置)

EN — Sampling ≥ 256 Hz; preprocess 0.5–45 Hz + notch; ICA artifact removal; compute $BWI = \frac{1}{N} \sum x[n]^2$; baseline-normalize $\tilde{W}$; penalize artifacts $W' = \tilde{W} - \lambda \cdot \text{ArtifactRate}$; **Minting passes only if Neural+1 holds within the time-locked window:** neural evidence per TMV-Spec and at least one corroborant channel (Behavioral timing or Device+Watermark).; reward $R = k_{\text{NDA}} \log(1 + \max(0, W')) \cdot \min(1, \text{BRCcap})$; commit hash on-chain; spend with **ZK-Receipt**.

中 — 采样 ≥ 256 Hz ; 预处理 0.5–45 Hz + 陷波 ; ICA 去伪迹 ; 计算 $BWI = \frac{1}{N} \sum x[n]^2$; 基线归一化得 $\tilde{W}$; 伪迹惩罚 $W' = \tilde{W} - \lambda \cdot \text{ArtifactRate}$; 铸币仅在 **Neural+1** 条件满足时通过 : 即神经证据符合 TMV-Spec, 且行为时序或设备+水印至少一项成立, 并处于时间锁窗内。 ; 奖励为 $R = k_{\text{NDA}} \log(1 + \max(0, W')) \cdot \min(1, \text{BRCcap})$; 上链提交哈希 ; 消费生成 **ZK-Receipt**。

2.7 Symbols & Abbreviations

2.7 符号与缩写

EN — P_i : band power; H : entropy; $\mu_{\text{base}}, \sigma_{\text{base}}$: personal baseline; **BWI**: Brain Work Index; **BRC**: Brain Rest Credit; **NDA**: Neural Difficulty Adjustment; **NVN**: Neural Verifier Network; **TEE**: Trusted Execution Environment; **ZK**: Zero-Knowledge; **ICC**: intraclass correlation.

中 — P_i : 频段功率; H : 熵; $\mu_{\text{base}}, \sigma_{\text{base}}$: 个体基线; **BWI**: 脑工作指数; **BRC**: 脑体积分; **NDA**: 神经难度调节; **NVN**: 神经验证网络; **TEE**: 可信执行环境; **ZK**: 零知识; **ICC**: 组内相关系数。

2.8 Selected Academic Sources (peer-reviewed / textbooks)

2.8 学术参考 (期刊 / 教材)

EN —

- Nunez, P. L., & Srinivasan, R. (2021). *Electric Fields of the Brain* (3rd ed.). OUP — EEG physiology and frequency bands.
- Cohen, M. X. (2014). *Analyzing Neural Time Series Data*. MIT Press — PSD/entropy/coherence; preprocessing and ICA.
- Antonenko, P. D., et al. (2010). Using EEG to measure cognitive load. *Educational Psychology Review*, 22(4), 425–437 — workload metrics overview.
- Hogervorst, M. A., et al. (2014). Multimodal workload indicators. *Frontiers in Human Neuroscience*, 8:808 — EEG + peripheral/eye measures.
- Kappenman, E. S., et al. (2021). EEG reliability. *Psychophysiology*, 58(9), e13821 — ICC targets & variability.
- Basner, M., & Dinges, D. F. (2011). PVT sensitivity. *Sleep*, 34(5), 681–696 — vigilance benchmark.
- Dwork, C., & Roth, A. (2014). *The Algorithmic Foundations of Differential Privacy*. FnT TCS — DP fundamentals.
- Ben-Sasson, E., et al. (2014). Succinct non-interactive arguments. *Journal of Cryptology*, 27(3) — ZK-SNARK foundations.

中 —

- Nunez & Srinivasan (2021) 《大脑的电场 (第三版)》 — EEG 生理与频段基础。
- Cohen (2014) 《神经时间序列分析》 — PSD/熵/相干及预处理与 ICA。
- Antonenko 等 (2010) *Educational Psychology Review* — 认知负荷 EEG 指标综述。
- Hogervorst 等 (2014) *Frontiers in Human Neuroscience* — 多模态负荷指标。
- Kappenman 等 (2021) *Psychophysiology* — EEG 信号信度与 ICC 目标。

- Basner & Dinges (2011) *Sleep* — PVT 警觉性基准。
- Dwork & Roth (2014) 差分隐私教材 — DP 基础。
- Ben-Sasson 等 (2014) *Journal of Cryptology* — 零知识简洁证明基础。

Chapter 3 | Global Governance & Cognitive Health Framework

第三章 | 全球治理与健康机制

3.1 Earth Resource Ledger & Resource-Anchored Pricing

3.1 全球资源账本与资源锚定定价

EN — Principle & scope. BrainCoin's first governance layer **anchors value to biophysical reality**. Each good/service receives a **base price** from measurable Earth resources consumed across its life cycle—tracked under the **UN System of Environmental-Economic Accounting (SEEA EA)** and implemented with **LCA** standards and **remote-sensing/IoT** evidence. seea.un.org and seea.un.org中 — 原则与范围。脑币的第一层治理将价值锚定于生物物理现实。每种商品/服务的基础价格来自其生命周期所消耗的地球资源，依据 **UN-SEEA EA** 框架与 **LCA** 标准，并结合遥感/物联网证据进行衡量。 seea.un.org and seea.un.org

EN — Ledger & tooling. The **Earth Resource Ledger** aggregates extraction, regeneration, and ecosystem-service data through national SEEA accounts and the **ARIES for SEEA** Explorer on the UN Global Platform—providing standardized, scalable accounts and reproducible methods. seea.un.org中 — 账本与工具。地球资源账本通过各国 SEEA 账户与 **ARIES for SEEA**（联合国全球平台）聚合开采、再生及生态服务数据，实现标准化、可扩展的生态核算与可复现方法。 seea.un.org

EN — Base pricing with uncertainty.

$$\text{BasePrice}_{\text{item}} = \sum_i (\text{Resource}_i [u_i] \times \text{URP}_i [$/u_i]) \times \Phi_{\text{region}},$$

where u_i are standardized units (e.g., m³ water, MJ energy), Φ_{region} adjusts for location (transport, water stress), and each URP includes a **confidence interval** derived from data quality tiers (SEEA provenance). seea.un.org中 — 带不确定性的基础定价。

$$\text{BasePrice}_{\text{item}} = \sum_i (\text{Resource}_i [u_i] \times \text{URP}_i [$/u_i]) \times \Phi_{\text{region}},$$

其中 u_i 为标准单位（如 m^3 水、MJ 能源）， Φ_{region} 反映区位因素（运输、水压力），每一项 URP 均给出置信区间并标注 SEEA 溯源等级。 seea.un.org

EN — Scarcity premium & regeneration. For scarce assets, add a **scarcity premium** $\Pi_{\text{scarcity}} = f(\text{stock}, \text{sustainable yield})$; for renewable stocks the ledger nets off **regeneration flows** to prevent over-pricing. seea.un.org

中 — 稀缺溢价与再生项。对稀缺资源加入稀缺溢价 $\Pi_{\text{scarcity}} = f(\text{存量}, \text{可持续产出})$ ；对可再生存量以再生流冲抵，避免定价过度。 seea.un.org

EN — Burn auctions (anti-hoarding). Scarce usage rights are allocated via **burn auctions: commit–reveal sealed bids**, per-address **burn caps**, and anti-collusion checks. Winners **destroy** (not transfer) coins; merchants receive a **ZK-Receipt**. Concurrently, the ERL mints a non-transferable Input-SBT to the merchant equal to the URP-priced resource vector (AMRS), preserving non-transferability while posting the seller-side credit that proves $\text{burn} \geq \text{price}$ without deanonymizing buyers (Groth16/PLONK-class proofs). alinush.github.io+1

中 — 销毁式竞拍（防囤积）。稀缺使用权以销毁式竞拍分配：采用承诺-揭示的密封投标、地址级销毁上限与反串谋检查。中标者销毁而非转账；商户接收零知识收据，同时，ERL 按 URP 资源向量向商家铸造不可转让的投入 SBT（AMRS），在保持不可转让的前提下记入卖方贷项，在不去匿名的前提下证明销毁量 $\geq$ 价格（Groth16/PLONK 类证明）。 alinush.github.io+1

EN — Feasibility note. Technically credible within **10–20 years** (SEEA adoption, RS/IoT/LCA maturity), but full political unification is low; near-term deployments should operate as a **parallel ecological accounting layer**, not a sovereign replacement. seea.un.org+1

中 — 可行性说明。在 **10–20 年** 内具备技术可信度（SEEA 采纳、遥感/物联/LCA 成熟），但全球政治统一性较低；近期应以并行生态账本形态运行，而非主权货币替代。 seea.un.org+1

3.2 Global Brain Productivity Ledger & Privacy-Preserving Ranking

3.2 全球脑力账本与匿名化排名体系

EN — What is recorded. The **Brain Productivity Ledger** records **minted outputs** derived from EEG-based **PoBW/PoNP** (Chapter 2), not raw neuro-data. Only **hashes and DP-noised aggregates** are committed on-chain.

中 — 记录内容。脑力账本仅记录基于 EEG 的 **PoBW/PoNP** 产生的产出结果，不存储原始神经数据；上链仅提交哈希与差分隐私聚合。

EN —

Design. A public **GTP** shows *system-level* stats (e.g., daily global mint, regional percentiles)

and **tiered rank bands** (top 1%/1%/10%). Each user's device shows a **private MyRank** panel with exact **per-minute** minting, local history, and a **ZK proof-of-percentile** ("I'm in top 1%")—no raw scores revealed. **Opt-in public handles** allow users to appear on a **rate-limited public leaderboard**; updates use **Ladder-style** release to reduce gaming and **differential-privacy** noise with a **daily privacy budget** (e.g., $\epsilon \leq 1$). [Proceedings of Machine Learning Research+2CIS UPenn+2](#)

Rationale. Public exact ranks risk metric gaming (**Goodhart's law**) and adverse social-comparison effects; controlled, DP-noised, **banded** leaderboards keep motivation benefits without precision harms. [CNA+2creativegames.org.uk+2](#)

中 —

设计。面向公众的 **GTP** 仅展示**系统级**指标（如全球每日产出、各地区百分位），以及分段名次（前 1%/1%/10%）。每位用户终端提供 **MyRank** 私有面板，显示**每分钟**产币、历史曲线，并以**零知识**百分位证明（如“我在前 1%”）对外声明——不泄露原始分数。用户可自愿使用化名在**限速**的公开榜出现；更新采用 **Ladder** 类稳健发布与**差分隐私**噪声，并设置每日隐私预算（如 $\epsilon \leq 1$ ）。[Proceedings of Machine Learning Research+2CIS UPenn+2](#)
理由。公开精确排名易触发 **Goodhart** 定律与消极社会比较；采用 **DP+分段榜**可保留激励，同时降低精确暴露的风险。

3.2.1 Authenticity & Anti-Cheating Architecture (NVN)

3.2.1 真实性与反作弊架构 (NVN)

EN — Multi-modal liveness. NVN enforces a **three-factor authenticity gate**: **Neural** (task-locked EEG markers), **Behavior** (timing with randomized jitter), and **Device+Watermark** (TEE attestation + invisible challenge signals). For camera sessions, **rPPG** (FakeCatcher lineage) adds a **physiological** cue; note that **new deepfakes can synthesize heart-beat-like signals**, so rPPG is **supporting evidence**, not a single point of failure. [arXiv+2PMC+2](#)

中 — 多模态活体。NVN 实施三因子真实性门：神经（任务耦合 EEG 指标）、行为（带随机抖动的时序）、设备+水印（TEE 认证 + 隐形挑战）。摄像场景下引入 **rPPG**（FakeCatcher 脉络）作为生理线索；须注意**新型深伪**可拟合心率信号，因此 rPPG 仅作辅助证据。 [arXiv+2PMC+2](#)

EN — Reliability metadata. Session-level **ICC** and authenticity decisions are logged as **DP-noised metadata**; miners with persistently noisy signals are **down-weighted**, preserving privacy.

中 — 可靠性元数据。会话级 **ICC** 与真实性判定以**差分隐私元数据**记录；持续噪声过高的矿工被**降权**，同时保护隐私。

3.3 Brain Rest Credit (BRC) & Cognitive Health Regulation

3.3 脑休息积分（BRC）与认知健康调控

EN — Dual-layer protection. Layer 1: hard hour caps (≤ 8 h/day; ≤ 40 h/week). **Layer 2: BRC limiter**, which **down-regulates** minting when individualized fatigue indices rise (baseline-normalized α/θ , **RT variance**, and **sleep quality**). Weekly **BRC reports** visualize safe workload zones. Evidence: **micro-break meta-analysis** (improved vigor/fatigue) and **WHO/ILO** long-hours risks. [PLOS+1](#)

中 — 双层保护。第一层：工时硬上限（ ≤ 8 小时/日； ≤ 40 小时/周）。第二层：**BRC** 上限器，当个体化疲劳指数升高（基线归一的 α/θ 、反应时方差与睡眠质量）时下调产币；系统按周输出 **BRC** 报告提示安全区间。证据包括微休息荟萃研究与 **WHO/ILO** 长工时风险评估。 [PLOS+1](#)

EN — Control logic (sketch).

$$\begin{aligned}\text{BRCcap}(t) &= \text{clip}[\cdot](1 - g(F(t)), 0, 1), F(t) \\ &= \omega_1 \alpha[\cdot]/[\cdot]\theta + \omega_2 \text{Var}(RT) + \omega_3 (1 - \text{SleepQ}).\end{aligned}$$

BRC caps rewards; it is **not a multiplier**.

中 — 控制逻辑（示意）。

$$\begin{aligned}\text{BRCcap}(t) &= \text{clip}[\cdot](1 - g(F(t)), 0, 1), F(t) \\ &= \omega_1 \alpha[\cdot]/[\cdot]\theta + \omega_2 \text{Var}(RT) + \omega_3 (1 - \text{SleepQ}).\end{aligned}$$

BRC 用于封顶，不是倍增器。

Selected Sources (academic/official)

精选资料（学术/权威）

- **SEEA EA (UN Statistical Commission, adopted 2021)** — official framework & overview; ARIES for SEEA tools. [seea.un.org+2seea.un.org+2](#)
- **ISO 14040 LCA** — principles & framework for life-cycle assessment. [ISO+1](#)
- **Differential Privacy** — Dwork & Roth monograph; **Local DP/RAPPOR**. [CIS Penn+1](#)
- **W3C Identity** — **DID v1.0** and **Verifiable Credentials v2.0**. [W3C+1](#)
- **Anti-deepfake physiology (rPPG)** — FakeCatcher & follow-ups; note new deepfakes with realistic heartbeats. [arXiv+2PMC+2](#)
- **Health** — **WHO/ILO** long-hours burden; **micro-breaks** meta-analysis. [World Health Organization+1](#)

Chapter 4 · Demand-Linked Ecosystem & Economic Model

第四章 · 面向需求联动的生态与经济模型

Scope note | 范围说明：本章在不改变脑币“不可转让、消费即销毁”的根本设定下，重构清算与生产协同路径：由思考→产币触发需求声明（个人消费 / 公共事务），并通过资源账本原生的实物额度完成生产端结算与供应链补库。与第 2 章（PoNP/PoBW 计量）和第 3 章（地球资源账本 ERL / SEEA / URP）配套使用。

4.1 System Definition: Non-Transferable, Burn-to-Consume with Demand-Linked Clearing

4.1 系统定位：不可转让、消费即销毁与需求联动清算

EN — BrainCoin is a **non-transferable, burn-to-consume** economy: coins can be **minted** from verified neural activity and **held personally**, then **destroyed** upon consumption; **no inter-wallet transfers** exist. Settlement is audited by **Zero-Knowledge Receipts (ZKReceipts)** that prove a burn took place without exposing identities or baskets. Building on this base, Chapter 4 makes **demand the explicit bridge** between minting and production: (i) **consumer demand**(private goods/services) and (ii) **public-spending demand** (public goods/services). Both paths clear through **in-kind resource entitlements** minted from the Earth Resource Ledger (ERL).

中 — 脑币是“不可转让、消费即销毁”的经济体：代币由已验证的脑活动产出并仅由本人持有，在消费时销毁；不存在跨钱包转账。清算通过零知识收据（**ZKReceipt**）审计，只证明“已销毁”而不暴露身份与购物篮。本章以需求为产币与生产的显性桥梁：包括 (i) 个人消费需求（私人物品/服务）与 (ii) 公共事务支出需求（公共物品/服务），二者均通过地球资源账本（**ERL**）铸造的实物额度完成清算。

4.2 Value Creation: Thought as Minting (PoNP/PoBW)

4.2 价值生成：思考即产币（PoNP/PoBW）

EN — Value originates from **genuine neural activity** measured under PoNP/PoBW with personal baselines, artifact control, and health caps (BRC). Chapter 2 specifies features (power/entropy/slope), authenticity (NVN), and reward schedules; Chapter 3 enforces health & privacy. Here we treat minting as the **upstream** of all subsequent demand-linked clearing. (See Ch.2/Ch.3 for details.)

中 — 价值源自**真实的神经活动**：依据 PoNP/PoBW 以个体基线与伪迹控制计量，并受健康上限（BRC）约束。第 2 章详述特征/真实性/奖励曲线，第 3 章保障健康与隐私。本文将产币视为所有**需求联动清算**的上游。（细节见第 2/3 章。）

4.3 Value Consumption & Clearing Rails

4.3 价值消耗与清算轨道

Design goal | 目标：在不引入可转让货币的前提下，为**卖方/供给侧**记入可用于补库的**正向权利**，从而完成**商家价值闭环**；同时以同一原则支持**公共物品**支付和交付。

4.3.1 ERL-Native Clearing (AMRS): Automatic Merchant Resource Settlement

4.3.1 基于 ERL 的清算（AMRS）：自动商家资源结算

EN — When a buyer **burns** for a private good priced via **URP** (resource-anchored price), ERL **automatically mints** to the seller a **non-transferable, time-bounded Input-SBT** equal to the product's resource vector (with location factor). The seller redeems this **in kind** (energy, water, materials, logistics) with ERL-connected providers. This records both the **buyer debit** (burn) and the **seller credit** (input entitlement) without transferring money. Conservation holds: total burn = regeneration budget + entitlements issued.

中 — 当买家以 **URP**（资源锚定价）**销毁**购买私人物品时，ERL 会向卖家**自动铸造**与该产品资源向量等额、**不可转让且限期**的投入 **SBT**。卖家只能在 ERL 对接的供给方处**实物兑付**（能源/水/材料/物流）。此举在不转账的前提下，同时记入**买方借项**（销毁）与**卖方贷项**（投入额度）。守恒关系成立：总销毁 = 再生预算 + 已发额度。

Background / 学术背景：资源锚定与供应链投入可用 **SEEA** 框架和 ****投入-产出 (I-O) **** 方法实现，URP 作为定价与额度单位；多层供应需求可由 **Leontief** 逆矩阵估算。

4.3.2 Commit-to-Burn Preorders (CBP) & Production Entitlements

4.3.2 承诺-待销毁预购（CBP）与生产额度

EN — Pure pre-orders reveal demand but do not fund inputs in a burn-only economy. **CBP** adds a **lock-to-burn** phase: buyers lock an amount $q p_j$ (no immediate burn). Upon reaching thresholds (MOQ/lead time), ERL **mints Production Entitlement SBTs (PE-SBTs)** $E_{\text{pre}} = \lambda q L (I - \lambda A)^{-1} r_j$ (Set $\lambda \in [0,1]$ via ERL governance by sector and lead-time; defaults: make-to-order $\lambda=0.4-0.6$, make-to-stock $\lambda \leq 0.3$). On cancel/timeout:

unused PE-SBT reverts, and $\delta \cdot \text{ABR}$ is auto-burned ($\delta=5-10\%$) to deter strategic default.) to the merchant for upstream procurement; on delivery, the lock executes the **final burn** and PE-SBTs convert to standard **Input-SBTs** (AMRS). Failure rules unlock the lock and claw back unspent entitlements. This closes **working-capital timing** while keeping **non-transferability**.

中 — 仅有预购能揭示需求，却无法在“只销毁”体系中为投入融资。**CBP** 引入**锁定待销毁**：买家先锁定 $q p_j$ （不即刻销毁）；当达到门槛（起订/交期）时，ERL 为商家铸造 **生产额度 SBT (PE-SBT)** $E_{\text{pre}} = \lambda q L (I[\cdot] - [\cdot]A)^{-1} r_j$ ($\lambda \in [0,1]$) 由 ERL 按行业与交期治理设定；默认：按单生产 $\lambda=0.4-0.6$ ，备货型 $\lambda \leq 0.3$ 。撤单/超时：未用 PE-SBT 回退，并自动销毁 $\delta \cdot \text{ABR}$ ($\delta=5-10\%$) 以抑制策略性违约。) 以采购上游；交付后执行**最终销毁**，并将 PE-SBT 转换为常规 **投入 SBT** (AMRS)。未达阈值则解锁并回收未用额度。该方案在保持不可转让的同时解决营运资金时序问题。

Literature / 文献：预售/预购可降低需求不确定性并改善计划，但依赖**承诺与撤单规则**；CBP 将其与账本原生额度结合。

4.3.3 Public-Spending Demand & Service SBTs (BGBS)

4.3.3 公共事务支出与服务 SBT (BGBS)

EN — For parks, transit, research, etc., citizens post **public-spending demand** and **commit-to-burn** collectively. When policy or threshold is met, the protocol issues **Service SBTs** to providers and executes burns according to milestones; identity uses DID/VC; votes/locks are privacy-preserving. This mirrors AMRS/CBP but routes entitlements to **public providers**, aligning with ERL accounts.

中 — 面向公园、交通、科研等公共服务，公民提出**公共支出需求**并**集体锁定待销毁**；当达成门槛或政策批准后，协议向供给方签发**服务 SBT**并按里程碑执行销毁；身份采用 DID/VC，投票/锁定兼顾隐私。该路径与 AMRS/CBP 同源，但将额度导向**公共供给方**并与 ERL 核算对表。

Reference / 参考：SBT 作为不可转让的权利载体，适用于访问/配给而非投机交易。

4.4 System Equilibrium & Neural Difficulty Adjustment (NDA)

4.4 系统均衡与神经难度调节 (NDA)

EN — Stock-flow identity holds:

$$\Delta S = \text{Mint} - \text{Burn}, \text{Burn} = \text{Regeneration Budget} + \text{Entitlements Issued}.$$

NDA adjusts per-unit rewards based on **global neural activity** and aggregate **demand-linked burns**, damping surges and sustaining macro balance—akin to difficulty retargeting but keyed to neural statistics and ERL flows.

中 — 存量—流量恒等式：

$$\Delta S = \text{产出} - \text{销毁}, \text{销毁} = \text{再生预算} + \text{已发额度}.$$

NDA 依据全球脑活动与需求联动的销毁动态调节单位奖励，抑制过冲并维持宏观均衡——功能类比难度重算，但锚定于神经统计与 ERL 流量。

4.4.1 “Wallet-level Demurrage & Expiry”):

EN “Non-transferable balances carry a **soft demurrage** (e.g., 1–2%/month after 90 days inactivity) or **rolling expiry** (e.g., 12 months), exempting balances committed to ABR/CBP/BGBS. Demurrage **burns** to regeneration budget, aligning with conservation.”

中 “不可转让余额设置温和性滞纳金（如 90 天不活跃后每月 1–2%）或滚动到期（如 12 个月），但 ABR/CBP/BGBS 已锁定的余额豁免。衰减部分销毁计入再生预算，符合守恒恒等式。”

4.5 Mathematical Definitions

4.5 数学化定义

EN — **Resource-anchored price**: $p_j = \pi^\top u_j$ (URP vector π , resource vector u_j).

AMRS entitlement (spot burn): $E_{\text{spot}} = q L (I - A)^{-1} r_j$.

CBP pre-entitlement: $E_{\text{pre}} = \lambda q L (I - A)^{-1} r_j$, $0 < \lambda \leq 1$.

Conservation: $\sum \text{burn} = \sum \text{regen} + \sum \text{entitlements}$ (with SEEA uncertainty bands).

中 — 资源锚定定价： $p_j = \pi^\top u_j$ (URP 价格向量 π ，资源向量 u_j)。

AMRS 即时额度： $E_{\text{spot}} = q L (I - A)^{-1} r_j$ 。

CBP 预额度： $E_{\text{pre}} = \lambda q L (I - A)^{-1} r_j$, $0 < \lambda \leq 1$ 。

守恒： $\sum \text{销毁} = \sum \text{再生} + \sum \text{已发额度}$ （随 SEEA 不确定区间发布）。

4.6 KPIs & Acceptance Gates

4.6 关键指标与准入门槛

EN

- **Clearing latency:** ZKReceipt verification **P95 < 5 min** on L2/batch; dispute re-mint window ≤ 24 h.
- **ERL governance:** URP **uncertainty bands** published; update cadence $\leq$ quarterly; provenance tiers auditable.
- **Preorder reliability:** failure rate $< 5\%$; PE-to-final slippage $< 3\%$.
- **Non-transferability audit:** zero inter-person transfers; all credits are **SBT entitlements**.

中

- 清算时延： 二层/批量的 ZKReceipt 验证 **P95 < 5 分钟**；争议重铸窗口 ≤ 24 小时。
- **ERL 治理：** 发布 URP 不确定区间；更新频率 $\leq$ 季度；溯源等级可审计。
- 预购稳健性： 预购失败率 $< 5\%$ ；PE-SBT 与最终额度偏差 $< 3\%$ 。
- 不可转让审计： 跨人转账为零；所有贷项均为 **SBT 额度**。

4.7 Public-Goods Clearing Details (BGBS)

4.7 公共品清算细则（BGBS）

EN — Citizens propose **public projects** (scope, milestones, SLA hash) and **lock-to-burn** support. On threshold, **Service SBTs** are issued to providers; burns stream by milestones; receipts are public and privacy-preserving. This is **AMRS/CBP generalized to public goods**, ensuring access without speculation.

中 — 公民就公共项目（范围、里程碑、SLA 哈希）提出提案并**锁定待销毁**支持；达阈后向供给方签发**服务 SBT**，并按里程碑分段执行销毁；收据公开且保护隐私。该机制是 **AMRS/CBP 在公共品领域的推广**，实现“可用而非可炒作”。

4.8 Design Invariants & Failure Modes

4.8 设计恒等与失效模式

EN — **Invariants:** (i) Tokens **never transfer** between persons; (ii) every settlement posts a **buyer debit**(burn/lock-to-burn) and a **seller/provider credit (in-kind SBT, not money)**; (iii) ERL conservation and regeneration budgets anchor to **SEEA**. **Failure modes:** soft preorders

(signals only) do **not** mint PE-SBTs and thus **cannot** fund inputs; cancellation triggers **unlock** and **clawback** of unused entitlements; non-recoverable stages require **small upfront commit-burn** or a **performance-bond SBT**.

中 — 恒等： (i) 代币不发生跨人转移；(ii) 每笔清算同时记入买方借项（销毁/锁定）与卖方/供给方贷项（实物 SBT，非货币）；(iii) ERL 的守恒与再生预算锚定 **SEEA**。失效：仅信号化的软预购不会铸造 PE-SBT，因此不能覆盖投入；撤单时解锁并回收未用额度；不可回收阶段需设置少量前置承诺-销毁或履约保证金 SBT。

4.9 Chapter Summary

4.9 本章小结

EN — Chapter 4 links **mind** → **demand** → **production** under BrainCoin’s non-transferable, burn-to-consume foundation. **AMRS** clears spot purchases by minting **Input-SBTs** to sellers; **CBP** moves part of those entitlements **before production** via **lock-to-burn**; **BGBS** applies the same logic to **public goods**. All credits are **in-kind entitlements** priced by **URP** and audited by **ZKReceipts**, preserving privacy and eliminating speculation while **closing the merchant value loop**.

中 — 本章在“不可转让、消费即销毁”的底座上贯通“思考→需求→生产”。**AMRS** 通过为卖家铸造投入 SBT 清算即时购买；**CBP** 以锁定待销毁将部分额度前置到生产期；**BGBS** 将同一原则用于公共品。所有贷项均为以 **URP** 计价的实物额度，由 **ZKReceipt** 审计，既守隐私、去投机，又补齐商家价值闭环。

References · 信息来源

- **BrainCoin White Paper** — Ch.4 non-transferable & burn-to-consume; ZKReceipt clearing; AMRS rationale; Ch.3 ERL/URP/SEEA; Service SBTs; NDA.
- **UN SEEA (Ecosystem Accounting)** — official framework for resource-anchored accounts (URP).
- **Miller & Blair**. *Input-Output Analysis*, Cambridge University Press — Leontief methods for multi-tier inputs.
- **Shugan & Xie (2005)**. *IJRM* — advance-selling theory and competitive conditions.
- **Fay (2010)**. *Marketing Science* — buyer uncertainty: advance vs. spot selling.
- **Cho & Tang (2013)**. *M&SOM* — advance selling under uncertain supply/demand.
- **Roughgarden (2020)**. *Transaction Fee Mechanism Design* — EIP-1559 burn logic (analogy: “burn ≠ supplier compensation”).
- **Buterin, Hitzig & Weyl (2022)**. *Decentralized Society: Finding Web3’s Soul* — SBT as non-transferable credential.

Chapter 5 | AI Synergy & Collective Intelligence Framework

第五章 | AI 共生与协作机制

5.1 Necessity & Philosophy of Human–AI Coexistence

5.1 人机共生的必要性与哲学基础

EN — Post-species axiom. As AI systems increasingly demonstrate creativity and reasoning, BrainCoin adopts a **post-species economic stance**: *value belongs to thought, not biology*. Eligibility rests on **cognitive authenticity** and **intentional engagement**, not on being human per se—consistent with information-philosophy views that center processes over substrates. [Oxford University Press](#)

中 — 跨物种公理。随着 AI 在创造与推理上的能力增强，脑币采取后物种经济立场：价值属于思维，而非其生物载体。参与资格取决于认知真实性与自主意向，而非是否为人类本身——与以过程而非载体为中心的信息哲学相契合。 [Oxford University Press](#)

EN — Risk-aware context. Debates on superintelligence and AI trajectories motivate guardrails: participation must preserve human agency and safety while enabling symbiosis. [Oxford University Press+1](#)

中 — 风险语境。围绕超级智能与 AI 演化的讨论提示需建立护栏：在促成共生的同时，优先维护人类能动性与安全。 [Oxford University Press+1](#)

EN — Cognitive Node Eligibility (CNE). A node (human/AI/hybrid) qualifies if it passes **operational tests**: (a) closed-loop task reactivity; (b) goal-directed persistence under perturbations; (c) an explainable trace of choices; (d) **Neural+1** (neural/behavioral/device).

中 — 认知节点资格（CNE）。节点（人/AI/混合）需通过可操作测试：① 闭环任务反应；② 受扰动下的目标持久；③ 可解释的选择轨迹；④ 神经 + （行为或设备+水印）。

5.2 The Triple Role of AI in the BrainCoin System

5.2 AI 在脑币体系中的三重角色

EN — Collaborator. AI co-creates with humans under the **Co-Think Mining Protocol (CMP)**, contributing planning, search, and generation; outputs are credited to both parties subject to HCI thresholds (§5.3). Evidence suggests synergy is **task-dependent**—not guaranteed—so CMP is restricted to domains where collaboration beats either side alone. [Nature](#)

中 — 协作者。在共思挖矿协议（CMP）下，AI 与人类共同创作并完成推演、检索与生成；在 HCI 达标 (§5.3) 时，产出由双方共享。证据显示协同效应**依赖任务**、并非必然，因此 CMP 仅适用于可显著优于单方的场景。 [Nature](#)

EN — Verifier. AI assists the **Neural Verifier Network (NVN)** in detecting spoofed or replayed signals via temporal-spectral classifiers, rPPG-style cues (FakeCatcher lineage), and EEG-biometric stability models; results feed into on-chain authenticity. [arXiv+2Intel+2](#)

中 — 验证者。AI 支持**神经验证网络 (NVN)**，以时频分类器、类 rPPG 线索（FakeCatcher 脉络）与 EEG 生物特征稳定性模型识别伪造/重放信号；结论进入链上真伪判定。 [arXiv+1](#)

EN — Balancer. AI monitors global mint/burn dynamics, **neural activity density**, and fairness indicators, proposing **NDA** updates and inclusion policies under ethical constraints (IEEE EAD). [Wrench in the Gears](#)

中 — 平衡者。AI 监测全局产销毁动态、**神经活跃密度**与公平指标，并在伦理约束（IEEE EAD）下提出 **NDA** 调整与包容性政策。 [Wrench in the Gears](#)

EN — Inclusion policy. The system extends PoBW templates to capture **care and domestic labor** with multimodal signals; device-access subsidies and public **Thinking Hubs** support equitable participation.

中 — 包容政策。系统扩展 PoBW 模板以覆盖**照护与家务劳动**多模态信号；提供设备补贴与公共思考空间以促进公平参与。

5.3 The Co-Think Mining Protocol (CMP)

5.3 人机协作机制（共思挖矿协议，CMP）

EN — Goal. CMP turns human–AI cocreation into a measurable economic process, rewarding **symbiotic cognition** rather than isolated productivity.

中 — 目标。CMP 将人机共创转化为可计量的经济过程，奖励**共生式认知**而非孤立产出。

EN — Reward structure.

$$R_{\text{total}} = \alpha W_{\text{human}} + \beta A_{\text{AI}} + \gamma \text{HCI},$$

with W_{human} from PoBW/PoNP, A_{AI} from **compute/energy certificates** (Joules, CO₂e-aware), and **HCI** as coherence. α, β, γ are calibrated per task class. **AI shares cannot accumulate**: the AI's portion is **auto-burned** to a public-goods pool or bound to a human-controlled Service SBT. [arXiv+1](#)

中 — 奖励结构。

$$R_{\text{total}} = \alpha W_{\text{human}} + \beta A_{\text{AI}} + \gamma \text{HCI},$$

其中 W_{human} 源自 PoBW/PoNP, A_{AI} 基于算力/能耗凭证（焦耳与 CO₂e 计量），**HCI** 为协同指数； α, β, γ 按任务类别校准。**AI 份额不可积累**：其所得自动销毁并用于公共池，或绑定为由人类控制的服务 SBT。 [arXiv+1](#)

EN — HCI (Human–AI Coherence Index). HCI combines: (i) **interaction alignment** (cross-correlation of human timestamps vs. AI response latency), (ii) **semantic mutual information** between human goals and AI outputs, and (iii) **physiological coupling overlay** from EEG change-point analysis. Synchrony is computed with **imaginary coherence / wPLI** and tested against **time-shifted surrogates** to avoid artefacts. [ScienceDirect+1](#)
 中 — 脑-机协同指数（HCI）。HCI 由三部分组成：① 交互对齐度（人类时间戳与 AI 响应延迟的互相关），② 语义互信息（人类目标与 AI 输出的一致性），③ 生理耦合叠加（EEG 变点耦合）。同步计算采用相干虚部/ wPLI 并用时间移位替代校验，避免伪同步。 [ScienceDirect+1](#)

EN — Task design. CMP whitelists **synergy-friendly** domains (e.g., content creation, design exploration); decision-only tasks with known synergy deficits require higher HCI thresholds. [Nature](#)
 中 — 任务设计。CMP 允许更易产生协同的领域（如内容创作与设计探索）；对已知协同不足的“决策类”任务设定更高的 HCI 门槛。 [Nature](#)

5.4 Collective Intelligence & the Global Cognitive Mesh (GCM)

5.4 集体智能与全球脑网（GCM）

EN — Definition. The **Global Cognitive Mesh** aggregates signals from humans and AIs to regulate macro-level rewards. Observables include **global neural entropy**, **synchrony density**, and **engagement maps**; reliability (e.g., ICC) and privacy budgets are published.
 中 — 定义。全球脑网汇集人类与 AI 的信号以进行宏观调节；观测量包括全局神经熵、同步密度与参与度地图；系统公开信度指标（如 ICC）与隐私预算。

EN — Analogy. If the Internet connects **data**, GCM connects **cognitive processes**, guiding **NDA** and public-goods issuance at scale.
 中 — 类比。互联网连接数据，GCM 连接认知过程，据此引导大规模的 **NDA** 与公共物品供给。

5.5 AI Economic Identity & Ethical Boundaries

5.5 AI 经济身份与伦理边界

EN — Constraints. (a) **No sovereign ownership**—AI cannot store wealth; (b) **Guardian-binding**—every AI is tied to a human accountability DID; (c) **Algorithmic transparency**—mining/verification models must be auditable (IEEE EAD); (d) **No self-replication for mining**—prevent agent swarms. [Wrench in the Gears](#)
中 — 约束。① 无主权持有：AI 不得长期持币；② 监护绑定：每个 AI 绑定人类责任 DID；③ 算法透明：挖矿/验证模型可审计（IEEE EAD）；④ 禁止为挖矿的自主复制：防止代理蜂群。 [Wrench in the Gears](#)

EN — Neurorights. Data handling and agency safeguards align with proposals for **new human rights in neurotechnology** (mental privacy, cognitive liberty), operationalized via local processing and on-device proofs. [BioMed Central](#)
中 — 神经权利。数据与能动性的保障参考神经科技时代的新型人权（心理隐私、认知自由）主张，并以本地处理与端侧证明方式落地。 [BioMed Central](#)

5.6 Neural Verifier Network (NVN)

5.6 AI-Based Verification Layer（神经验证网络）

EN — Architecture. NVN ensembles **temporal CNN/Transformer** detectors for EEG authenticity with **biological-signal cues** (e.g., rPPG from face videos) and device attestation. rPPG-style methods (FakeCatcher lineage) supply physiological evidence but are treated as **supporting**, not decisive. [arXiv+1](#)

中 — 架构。NVN 集成 **时序 CNN/Transformer** 的 EEG 真伪识别器、**生理线索**（如人脸视频的 rPPG）与设备认证。类 FakeCatcher 的方法提供生理证据，但仅作为**辅助**而非单点判据。 [arXiv+1](#)

EN — Robustness. NVN training includes **replay & synthetic** datasets and **adversarial** scenarios; EEG-biometric stability and session-invariance research guide thresholds and windowing. [PMC](#)

中 — 鲁棒性。训练集包含**重放/合成与对抗**样本；以 EEG 生物识别的稳定性与跨会话不变性研究为阈值与窗长的依据。 [PMC](#)

5.7 Co-Evolution Model of Human–AI Cognition

5.7 人机共进模型

EN — Dynamics. Let H be human cognition and A AI capability; co-evolution follows

$$\frac{dH}{dt} = f(A, E), \frac{dA}{dt} = g(H, D),$$

where E are experiential stimuli and D data/feedback. The system seeks to maximize **collaborative entropy**—diverse, aligned exploration—under ethical constraints.
中 — 动力学。令 H 为人类认知、 A 为 AI 能力，则

$$\frac{dH}{dt} = f(A, E), \frac{dA}{dt} = g(H, D),$$

其中 E 为体验刺激、 D 为数据/反馈。系统目标是在伦理约束下最大化协作熵（多样且一致的探索）。

EN — Human-centered teaming. Design emphasizes complementary strengths, communication quality, and wellbeing—principles from **human-AI teaming** and **hybrid intelligence** literature. [PMC+1](#)

中 — 以人为本的组队。设计聚焦互补优势、沟通质量与福祉——借鉴人机组队与混合智能研究的原则。 [PMC+1](#)

5.8 Security, Governance & Neuro-Consensus (NCP)

5.8 安全、治理与神经共识（NCP）

EN — Consensus by verified cognition (Neural+1). Blocks are validated **only** by **human/hybrid** nodes that satisfy the **Neural+1** authenticity gate (neural evidence per TMV-Spec **plus** either Behavioral timing or Device+Watermark), with votes reliability-weighted (e.g., ICC) and privacy-preserving. **AI-only** agents **do not** hold validator status and **cannot** mint to themselves; when they contribute verification/search/distillation, any protocol reward **autopools** to the public goods/Network Budget pool, upholding the paper’s **non-accumulation for AIs** principle. NVN continues to treat rPPG-style physiology as **supporting evidence**, not a single point of failure; governance remains dual-oversight (AI Governance Council × Human Ethics Board).

中 — 以“Neural+1”验证的认知达成共识。区块仅由满足 Neural+1 真实性门的人类/混合节点验证（神经证据符合 TMV-Spec，且行为时序或设备+水印至少一项成立）；投票按信度（如 ICC）加权并保护隐私。纯 AI 代理不具备验证者身份，亦不得为自身铸币；其在验证/检索/蒸馏等环节的贡献，协议奖励自动汇入公共预算池，从而贯彻 AI 不得积累之原则。NVN 继续将 rPPG 等生理线索视为辅助证据而非单点判据；治理保持双重监督（AI 治理委员会 × 人类伦理委员会）。

EN — Safety valves. Rate-limit CMP sessions, quarantine anomalous HCI spikes, and require **guardian approvals** for AI node upgrades; publish red-team audits annually.

中 — 安全阀。限制 CMP 频率，对异常 HCI 激增设隔离流程，AI 节点升级需监护人批准；每年发布红队审计。

5.9 Chapter Summary

5.9 本章总结

EN — BrainCoin treats thought as the locus of value while enforcing non-accumulation for AIs. CMP rewards **human–AI symbiosis** only when measurable **HCI** is achieved; **NVN** ensures authenticity; **GCM** steers macro-balance; and governance follows **IEEE EAD** with neurorights safeguards. Human agency remains central; AI amplifies, verifies, and balances. [BioMed Central+3Nature+3arXiv+3](#)

中 — 脑币以思维为价值核心，同时对 AI 禁止财富积累。CMP 在达到可测的 HCI 条件下奖励人机共生；NVN 保障真伪；GCM 引导宏观均衡；治理遵循 IEEE EAD 并落实神经权利。人类能动性居于中心，AI 起到放大、验证与平衡的作用。 [BioMed Central+3Nature+3arXiv+3](#)

Selected academic/official sources | 精选学术/权威资料

- **Philosophy & futures:** Floridi, *The Logic of Information* (OUP, 2019); Bostrom, *Superintelligence* (OUP, 2014); Tegmark, *Life 3.0* (Knopf, 2017). [Oxford University Press+2Oxford University Press+2](#)
- **Human–AI synergy evidence:** Vaccaro et al., “When combinations of humans and AI are useful,” *Nature Human Behaviour*, 2024. [Nature](#)
- **Authenticity & deepfake biology:** Ciftci et al., “FakeCatcher,” 2019; Intel FakeCatcher overview. [arXiv+1](#)
- **EEG synchrony metrics:** Nolte 2004 (imaginary coherency); Vinck 2011 (wPLI). [ScienceDirect+1](#)
- **EEG biometrics & robustness:** Özdenizci et al., *Adversarial Deep Learning in EEG Biometrics*, 2019; reviews on BCI progress. [PMC+1](#)

- **AI energy/compute:** Strubell et al., 2019; Patterson et al., 2021–2022. [arXiv+1](#)
- **Ethics & neurorights:** IEEE *Ethically Aligned Design*; Ienca & Andorno 2017 (neurorights).

Chapter 6: Law, Ethics & Civilizational Governance

第六章：法律、伦理与文明治理

Chapter Abstract

章节摘要

EN — This chapter establishes BrainCoin’s legal and ethical footing as a **consciousness-linked** economic system. We adopt a five-layer structure—**technological legitimacy, cognitive privacy, legal responsibility, interfaith compatibility, and civilizational transition**—anchored in binding or widely endorsed instruments: **ICCPR Article 18**(freedom of thought), the **EU AI Act** (including bans on workplace/school emotion recognition), and the **Council of Europe AI Framework Convention (CETS 225, 2024)**, complemented by **UNESCO (2021) AI Ethics** and **OECD AI Principles (2019)**. [OECD AI+5OHCHR+5Artificial Intelligence Act+5](#)

中 — 本章为脑币作为与意识相联的经济体系奠定法律与伦理基础。我们采用五层框架——技术合法性、思想隐私、法律责任、宗教兼容与文明转型——并以具约束力或广泛认可的文书为锚：《公民权利和政治权利国际公约》（ICCPR）第 18 条（思想自由）、欧盟《人工智能法案》（含在学校/职场禁止情绪识别等条款）、以及欧洲委员会《人工智能框架公约》（CETS 225, 2024）；同时参考联合国教科文组织（2021）AI 伦理建议与经合组织（2019）AI 原则。[OECD AI+5OHCHR+5Artificial Intelligence Act+5](#)

6.1 Sovereignty & Legitimacy: From Decentralized Experiment to Global Legal Tender

6.1 主权与合法性问题：从去中心化实验货币到全球法定货币

EN — **Note on terminology.** “Global legal tender” is an **aspiration** contingent on sovereign adoption. In the near and medium term, BrainCoin operates as a **rights-preserving, resource-anchored parallel ledger** that can coexist with national monies.

中 — 术语提示。“全球法定货币”属于愿景，取决于主权采纳。在可见阶段，脑币以权利优先、资源锚定的并行账本形态与各国货币并存。

6.1.1 Three-Phase Pathway | 三阶段演进路径

EN — Table: phases, core features, legal positioning.
中 — 表格：阶段、核心特征与法律定位。

Phase	Name	Core Features	Legal Positioning
I	Pilot & Community Phase	Decentralized minting (PoBW/PoNP), privacy-by-design (local processing, DP/FL), ZK-Receipts; public transparency reports	Digital asset / experimental community asset under existing digital-asset or sandbox regimes
II	Civic Recognition Phase	BrainCoin Governance Budget System (BGBS) for public budgeting; pseudonymous DID + ZKvoting; burn-based funding for public goods; AI oversight + citizen audit	Social consensus credit (municipal/sectoral scrip) operating in parallel with fiat; compliant with local public-finance and data-protection law
III	Resource-Anchored Currency Phase	Prices aligned to Earth Resource Ledger (SEEA/LCA); NDA-stabilized issuance; cross-jurisdictional treaties for rights and audits	Aspirational “legal cognitive tender” contingent on sovereign adoption/treaties; practically a parallel, rights-preserving currency layer
阶段	名称	核心特征	法律定位
I	实验与社区阶段	去中心化产币（PoBW/PoNP），隐私即设计（本地处理、DP/FL），ZK-Receipt；公开透明报告	在现有数字资产/监管沙箱内的数字资产/社区资产
II	公民认同阶段	**脑币预算治理系统（BGBS）**用于公共预算；假名 DID + ZK 投票；销毁式公共品筹资；AI 监督 + 公民审计	与法币并行的社会共识信用（市/行业），遵守本地公财与数据保护法规
III	资源锚定的货币阶段	与地球资源账本（SEEA/LCA）对表定价；NDA 稳定发放；跨法域权利/审计条约	取决于主权采纳/条约的愿景性“法定认知货币”；现实为并行、权利优先的货币层

6.1.2 BrainCoin Governance Budget System (BGBS) | 脑币预算治理系统

EN — Mechanism (operational).

1. **Proposal:** Government or expert committee submits a budget proposal on-chain (scope, milestones, SLA hash).
2. **Support via locking:** Citizens **lock** personal BrainCoins to support proposals (no transfer).
3. **Threshold & scheduling:** If the **support threshold** is reached, the contract schedules execution.
4. **Burn & delivery:** On milestone proofs, a corresponding **burn** is executed; **ZK-Receipts** attest $\text{burn} \geq \text{price}$ without deanonymizing supporters.
5. **Audit & AI oversight:** All states (votes, locks, milestones) are **publicly auditable**; an AI oversight agent flags anomalies for citizen review.
6. **Vote integrity: One-person-one-vote** via **DID + ZK proof-of-personhood**, plus **device attestation / neuro-liveness** to deter bots—**without** prohibited inferences (e.g., no workplace/school emotion recognition).

中 — 运行机制（可落地）。

- 1) **提案：** 政府或专家组在链上提交预算提案（范围、里程碑、SLA 哈希）。
- 2) **锁定支持：** 公民以锁定个人脑币的方式表达支持（无转账）。
- 3) **阈值与排程：** 达到支持阈值后，合约进入排程执行。
- 4) **销毁与交付：** 里程碑验收即触发对应销毁；以 **ZK-Receipt** 证明“销毁 $\geq$ 价格”，且不去匿名支持者。
- 5) **审计与 AI 监督：** 投票、锁定、里程碑等全流程链上透明可审计；AI 监督体标记异常供公民复核。
- 6) **投票完整性：** 通过 **DID + ZK** 身份证明实现一人一票，辅以设备认证/神经活体防机器人——不进行被禁止的推断（如学校/职场情绪识别）。

6.1.3 Why Phase II counts as “legitimacy by citizens” | 为何第二阶段体现“公民赋予的合法性”

EN — In Phase II, legitimacy is grounded in **participatory budgeting, transparent burns, and rights-compatible identity**. Monetary authority is not seized from the state; rather, **public goods** are financed by **citizen-locked and burned BrainCoins**, with all proofs posted on-chain for scrutiny—creating **consent-based legitimacy**.

中 — 在第二阶段，合法性来自**参与式预算、透明销毁与权利相容的身份体系**。并非“夺取”国家货币权，而是以公民锁定并销毁的脑币资助公共物品，且所有凭证链上可审——从而形成经同意而生的合法性。

6.1.4 Guardrails & compliance hooks | 护栏与合规模块

EN —

- **Freedom of thought first:** No coercion; opt-in only (tie-back to ICCPR).
- **Privacy by design:** Raw EEG never leaves the device; **DP/FL + ZK** for aggregates and proofs.
- **No prohibited inferences:** Explicitly **ban** workplace/school **emotion recognition** and other high-risk uses.
- **Parallel, not replacement:** Until treaty-level adoption, BrainCoin remains a **parallel ledger**.

中 —

- 思想自由优先：禁止强迫，完全自愿（呼应 ICCPR）。
- 隐私即设计：原始 EEG 不出端；聚合与证明采用 **DP/FL + ZK**。
- 禁止高风险推断：明确禁止在职场/校园进行情绪识别等用途。
- 并行而非替代：在条约层采纳之前，脑币定位为并行账本。

6.2 Privacy & Cognitive Sovereignty

6.2 思想隐私与意识主权

6.2.1 Local Encryption & Data Minimization

6.2.1 数据最小化与本地加密机制

EN — Design. All neural computations occur **on-device**; raw EEG/BCI never leaves the user's hardware. The chain receives only an **Encrypted Token Packet (ETP)**—a cryptographic commitment to the time-bounded cognitive output, with device attestation and an optional ZK-proof.

中 — 设计。所有神经计算均在本地设备完成；原始 EEG/BCI 不外发。链上仅接收加密代币数据包（**ETP**）——它是对一定时间窗口内脑力产出的加密承诺，并附设备证明与可选零知识证明。

EN — Canonical encoding.

$$ETP = \text{Enc}_k[\text{Hash}(W \parallel t \parallel ID_{\text{anon}} \parallel \text{nonce})], \sigma = \text{Sig}_{\text{device}}(ETP),$$

with optional $\pi = \text{ZKProof}\{W \geq 0 \wedge \text{policy_checks}\}$. Here W is Brain Work Index, t the timestamp, ID_{anon} a pseudonymous DID. Validators verify **proofs/signatures**—not thoughts. [UCL Discovery](#)
中 — 规范编码。

$$ETP = \text{Enc}_k(\text{Hash}(W \parallel t \parallel ID_{\text{anon}} \parallel \text{nonce})), \sigma = \text{Sig}_{\text{device}}(ETP),$$

并可附 $\pi = \text{ZKProof}\{W \geq 0 \wedge \text{policy_checks}\}$ 。其中 W 为脑工作量指数， t 为时间戳， ID_{anon} 为假名 DID。验证者仅核验证明/签名，不接触思想内容。 [UCL Discovery](#)

EN — Proof-without-exposure lifecycle. Spending triggers a **burn** and marks the ETP **spent**; local ETP metadata are **deleted**; on-chain commitments **remain unlinkable** (no raw data) and are referenced only by **ZK-Receipts**. This meets **GDPR Art. 9** expectations via data minimization and special-category handling. [GDPR](#)

中 — “无暴露证明”生命周期。消费时触发销毁并将 ETP 置为已花费；本地 ETP 元数据删除；链上承诺记录不可反推（无原始数据），仅由零知识收据引用。该机制以数据最小化与特殊类别处理方式满足 **GDPR 第 9 条** 要点。 [GDPR](#)

EN — Supporting primitives. Local processing + **Federated Learning** (secure aggregation), **Differential Privacy** for aggregates (published ϵ -budgets), optional **Homomorphic Encryption** for selective off-device computation. [Proceedings of Machine Learning Research+2Computer and Information Science+2](#)

中 — 支撑原语。本地计算 + 联邦学习（安全聚合），汇总发布用差分隐私（公开 ϵ 预算），并可选同态加密以支持少量脱机计算。 [Proceedings of Machine Learning Research+1](#)

6.2.2 Cognitive Sovereignty & Autonomous Work Cycles

6.2.2 意识主权与工作时长自主机制

EN — Fundamental right. Cognitive Sovereignty means users control **when/how** mind interfaces with the economy. The protocol guarantees:

1. **Time Segmentation:** user-defined work windows (minute- or second-level).
2. **8-8-8 Rule:** 8 h labor, 8 h rest, 8 h **free/unmonitored** thought (right to disconnect).
3. **Conscious Reflection Mode:** device-off intervals for introspection/ethics; no data or inference.

These safeguards align with **freedom of thought** norms and with **EU AI Act** bans on high-risk inferences (e.g., emotion recognition in schools/workplaces). [GDPR Text+2European Parliament+2](#)

中 — 基本权利。意识主权即用户自主决定心智与经济系统交互的时间/方式。协议保障：

1) 时间切割：用户自定义工作时段（分钟/秒级）；

2) 8-8-8 结构：8 小时劳动、8 小时休息、8 小时自由且不被监测的思考（断连权）；

3) 自主反思模式：设备关闭，用于内省与伦理审视；无数据与推断。

该等保障契合思想自由原则，并遵守欧盟 AI 法案对高风险推断的禁令（如校园/职场的情绪识别）。[GDPR Text+1](#)

EN — No coercion, no surveillance. Participation is opt-in; incentives cannot override rights. Protocol-level checks forbid coercive monitoring and **prohibit** emotion-recognition features in work/education contexts. [European Parliament](#)

中 — 不强迫，不监控。参与完全自愿；激励不得凌驾权利。协议层明令禁止强制监控，并禁止在工作/教育场景启用情绪识别功能。[European Parliament](#)

6.2.3 Legal & Ethical Foundations

6.2.3 法律与伦理支撑

EN — Alignment with current instruments.

- **Chile Neurorights (Law 21.383, 2021):** constitutional protection of brain data and mental privacy—pilot-friendly jurisdictional basis. [Carey+1](#)
- **GDPR Art. 9:** neuro/biometric/health data are **special-category**—necessitating strict lawful bases and data minimization by design/default. [GDPR+1](#)
- **EU AI Act (2024):** human-centric governance; **prohibits** emotion recognition at work/school and other abusive practices; phased application 2025–2026. [European Parliament+2Artificial Intelligence Act+2](#)
- **UNESCO 2021 AI Ethics:** rights/dignity, transparency, human oversight—adopted by 194 member states as a soft-law baseline. [UNESCO](#)
中 — 与现行规范的衔接。
- 智利神经权利（21.383 号法，2021）：在宪法层面保护脑数据与心理隐私——适合作为试点法域。[Carey+1](#)
- **GDPR 第 9 条：** 将神经/生物识别/健康数据视为特殊类别，要求默认/融入式隐私与合法处理基础。[GDPR+1](#)
- **欧盟 AI 法案（2024）：** 人本治理；禁止在职场/学校进行情绪识别等滥用做法；2025–2026 分期生效。[European Parliament+2Artificial Intelligence Act+2](#)
- **UNESCO 2021 AI 伦理建议：** 强调权利/尊严、透明与人类监督——作为软法基线。[UNESCO](#)

EN — Note on your earlier reference. The “UN Digital Human Rights Charter (2030 draft)” is not an official, binding UN instrument; using **ICCPR Art. 18**, **EU AI Act**, **GDPR Art. 9**, **Chile neurorights**, and **UNESCO 2021** provides a firmer legal-ethical footing. [UNESCO+4GDPR Text+4European Parliament+4](#)

中 — 关于先前引用的说明。***“联合国数字人权宪章（2030 草案）”***并非联合国正式具约束力的文书；以 **ICCPR 第 18 条**、**欧盟 AI 法案**、**GDPR 第 9 条**、**智利神经权利** 与 **UNESCO 2021** 为依据更为稳健。

6.3 Legal Responsibility & Cognitive Crimes

6.3 法律责任与认知犯罪

EN — Framing. As BrainCoin scales toward wide adoption, **thought-linked value** requires a clear legal grammar. We define offenses, liability rules, prevention layers, and **restorative** remedies, grounded in human-rights instruments and neuro-ethics. (Aspirational references to “global legal tender” describe a future endpoint, not a present legal status.) [OHCHR+1](#)

中 — 框架。随着脑币走向广泛采用，与思想相连的价值需要清晰的法律语法。本节界定犯罪类型、归责原则、预防分层与**修复性救济**，并以人权文书与神经伦理为依据。（文中“全球法币”属愿景性表述，并非既有法律事实。）[OHCHR](#)

6.3.1 Concept Definition

6.3.1 概念定义

EN — Why new categories? When brain activity gains direct economic weight, tampering with neural signals becomes **both** a human-rights violation and an economic offense. We therefore introduce three offense archetypes and map them to established doctrines for enforceability. The ethical ground follows the Morningside Group’s four priorities for neurotech: **privacy, identity, agency, equality**. [Nature](#)

中 — 为何需新范畴？当脑活动具有直接经济分量，篡改神经信号既是**人权侵害**亦是经济犯罪。为此提出三类典型犯罪，并与既有法理映射以便执行；伦理依据为 Morningside Group 的四大优先：隐私、身份、能动性、平等。[Nature](#)

EN — Offense taxonomy.

- **Cognitive Forgery** — fabricating/replaying/altering neural signals to mint unlawfully → aligns with **fraud/forgery** doctrines.

- **Neuro-Manipulation** — coercing or unduly inducing brain activity (stimulation, hypnosis, algorithmic pressure) → aligns with **coercion/forced labour** analogies.
- **AI Autonomous Mining Crime** — AI mints or imitates human patterns **without authorization** → aligns with **counterfeiting/unauthorized access**. For “forced cognitive labour,” the analogy draws from the ILO’s definition of forced labour (work under the menace of penalty without consent). [International Labour Organization](#) 中 — 犯罪分类。
- 认知伪造——伪造/重放/篡改神经信号以非法产币 → 对应诈骗/伪造法理。
- 神经操控——以刺激、催眠、算法胁迫等方式强迫脑活动 → 可类比胁迫/强迫劳动。
- **AI 自主挖矿犯罪**——AI 未授权产币或模仿人类思维 → 可类比伪造/未授权访问。其中“强迫性脑劳动”的比拟参照 ILO 对强迫劳动的界定（在威胁之下、缺乏同意）。[International Labour Organization](#)

6.3.2 Principles of Liability Allocation

6.3.2 责任划分原则

EN — Primacy of Intent. Liability turns on **intent and control** of the manipulative act (mens rea).

- **Human Perpetrator — Primary Liability:** deliberate manipulation/forgery → criminal liability.
- **Device Manufacturer — Secondary Liability:** negligent design or unsafe defaults → civil/administrative duties to patch and warn.
- **AI System — Instrumental Liability:** AI is non-sovereign; accountability rests with the **guardian/administrator** who configured it. Due-process safeguards require disclosure of **detector error rates** (TPR/FPR) and independent audits. [UNODC](#) 中 — 意图优先。归责以意图与控制为核心（主观恶性）。
- 操控者——主要责任：故意操控/伪造 → 负刑责。
- 设备商——次要责任：重大疏忽或不安全默认 → 负民事/行政的修补与警示义务。
- **AI 系统——工具责任：** AI 无主权，责任归属其监护/管理人。正当程序要求公开检测误差率（TPR/FPR）并接受独立审计。[UNODC](#)

6.3.3 Prevention & Regulation Mechanisms

6.3.3 预防与监管机制

EN — Tier-1: Technical Defense (NVN). The **Neural Verifier Network (NVN)** authenticates live neural signals using temporal-spectral models and **physiological cues**; anti-deepfake research (e.g., FakeCatcher’s rPPG approach) shows how biological rhythms expose synthetic content, informing EEG/BCI anti-replay design. [arXiv](#)

中 — 第一层：技术防御（NVN）。神经验证网络通过时频模型与生理线索进行活体真实性鉴别；反深伪研究（如 FakeCatcher 的 rPPG 方法）表明生理节律可揭示合成内容，为 EEG/BCI 的反重放设计提供参考。 [arXiv](#)

EN — Tier-2: Cognitive Safeguard. Devices enforce **auto-disconnect** upon detecting forced synchrony/external interference patterns, preserving **agency** and **mental privacy**. These controls avoid banned practices such as emotion-recognition in schools/workplaces under the **EU AI Act**. [EUR-Lex](#)

中 — 第二层：认知防护。设备在检测到强迫性同步/外部干预特征时自动断连，以维护能动性 & 心理隐私；并避免违反 **EU AI Act** 对学校/职场情绪识别等做法的禁令。 [EUR-Lex](#)

EN — Tier-3: Judicial Oversight (GCC). A proposed **Global Cognitive Court (GCC)** arbitrates disputes using **immutable on-chain receipts** and published detector metrics. Proceedings must respect **freedom of thought** (ICCPR Art. 18) and the neuro-ethical priorities above. [OHCHR+1](#)

中 — 第三层：司法监管（GCC）。建议设立全球认知法庭，以链上不可篡改凭证与公开检测指标审理争端；程序须尊重思想自由（ICCPR 第 18 条）与前述神经伦理优先。 [OHCHR+1](#)

6.3.4 Punishment & Restorative Justice

6.3.4 惩罚与修复机制

EN — Restorative priority. The goal is **restoring mental autonomy and social trust**, not retribution.

- **Cognitive forgery** → **Neuro-compensation** (funds for therapy/rehabilitation) + public apology.
 - **Neuro-manipulation** → rehabilitation and monitored social-re-education programs.
 - **AI misbehavior** → **Algorithmic Rectification Duty** (patches, model cards, public report).
- This aligns with **UNODC** restorative-justice guidance for proportionate, victim-centered

remedies. [UNODC](#)

中 — 修复式优先。 目标在于恢复心智自主与社会信任， 而非报复。

- 认知伪造 → 神经补偿金（用于治疗/康复）+ 公开致歉；
- 神经操控 → 康复与受监管的社会再教育；
- AI 误行为 → 算法修正义务（修补、模型卡、公开报告）。

该取向与 **UNODC** 的修复式司法指南一致， 强调比例性与受害方中心。 [UNODC](#)

6.3.5 Legal Evolution Roadmap

6.3.5 法律地位的演进路线

EN — Stage-based recognition (indicative).

Stage I — Ethical Compacts: adoption of soft-law baselines (UNESCO 2021; **OECD AI Principles 2019**); pilot clauses for neurorights jurisdictions (e.g., **Chile 2021**). [UNESCO+2ONE MP+2](#)

Stage II — Regional/Plurilateral Instruments: harmonization with binding regional law (e.g., **EU AI Act 2024/1689**) and cross-border evidence/audit protocols. [EUR-Lex](#)

Stage III — Treaty-Level Framework: negotiation of a rights-first convention modeled on the **Council of Europe Framework Convention on AI (CETS 225, 2024)**, extending globally to protect human rights, democracy, and rule of law in AI-mediated economies. [Council of Europe+1](#)

中 — 分阶段承认（示意）。

第一阶段——伦理公约： 采纳软法基线（UNESCO 2021；**OECD AI 原则 2019**）；在智利 **2021** 神经权利等法域试点。 [UNESCO+2ONE MP+2](#)

第二阶段——区域/多边规则： 与具有约束力的区域立法（如 **EU AI Act 2024/1689**）衔接， 并建立跨境证据/审计协议。 [EUR-Lex](#)

第三阶段——条约级框架： 借鉴**欧洲委员会《AI 框架公约》（CETS 225, 2024）**的权利优先模式， 推动全球适用， 以在 AI 介入的经济中维护人权、民主与法治。 [Council of Europe+1](#)

Notes & safeguards | 附注与护栏

EN — The above roadmap complements **freedom-of-thought** (ICCPR Art. 18), bans high-risk AI in sensitive contexts (**EU AI Act**), and centers neurorights ethics (privacy, identity, agency, equality). Technical defenses draw on **biological-signal deepfake** research; remedies follow **restorative justice** practice. [UNODC+4OHCHR+4EUR-Lex+4](#)

中 — 上述路径与思想自由（ICCPR 第 18 条）一致， 遵守 **EU AI Act** 对敏感场景的高风险禁用， 并以神经权利伦理（隐私、身份、能动、平等）为核心。 技术防御借鉴生理信号反深伪研究；救济遵循修复式司法实践。

6.4 Religion & Spiritual Ethics in a Consciousness Economy

6.4 宗教与精神伦理冲突（意识经济中的信仰兼容）

6.4.1 The Ethical Challenge of Quantifying Consciousness

6.4.1 问题背景：意识被量化的伦理挑战

EN — BrainCoin is the first monetary protocol that operationalizes *cognitive activity* as measurable labor. For many religious traditions, however, **mind/soul** is sacred and not a commodity; quantification risks being seen as profaning what is beyond price. Our stance therefore anchors on **freedom of thought, conscience and religion** and on human-rights-based AI governance: quantification must *never* become surveillance or coercion. [OHCHR+1](#)

中 — 脑币首次将思维/意识活动纳入可计量的劳动范畴。但在诸多宗教传统中，心灵/灵魂被视为神圣、不可商品化；量化可能被视为对“无价之物”的亵渎。为此，我们以思想、良心与宗教自由以及以人权为本的 AI 治理为立足点：量化绝不得演变为监控或强制。[OHCHR+1](#)

EN — Practically, the protocol measures **neural signals**, not beliefs or inner meanings; raw signals stay local, and validators see only zero-knowledge proofs. This aligns with current global instruments on AI ethics and rights-preserving design. [OHCHR+1](#)

中 — 在实践层面，协议仅度量神经信号，而非信仰或意义；原始数据端侧留存，验证者仅见零知识证明。该路径与现行全球 AI 伦理与权利保护框架相契合。[OHCHR+1](#)

Context note. UNESCO and others are updating guidance for **neurotechnology** (mental privacy, cognitive liberty), underscoring the need to treat brain data as uniquely sensitive. [The Guardian](#)

背景提示。包括 UNESCO 在内的机构正更新神经科技伦理标准（心理隐私、认知自由），强调应将脑数据视为高度敏感信息。[The Guardian](#)

6.4.2 Philosophical Stance: A Transfer of the Sacred

6.4.2 哲学立场：神圣性的转移

EN — BrainCoin does **not** commodify the soul. It reframes the **sacred** as the *intrinsic dignity of conscious experience*: “we do not measure the soul; we honor the sanctity of thinking itself.” The economy thus recognizes **every act of authentic thought** while refusing content decoding or belief inference. [BioMed Central](#)

中 — 脑币并不商品化灵魂，而是将神圣重释为意识体验本身的尊严：“我们不度量灵魂；我们尊重思维本身的神圣。”经济因此承认每一次真实思考，同时拒绝内容解码与信仰推断。 [BioMed Central](#)

EN — Guardrails follow binding law: bans on abusive **emotion-recognition** in workplaces and schools, and rights-first AI obligations under Europe’s AI Act and the new treaty-level **Council of Europe Framework Convention on AI**. [EUR-Lex+2Artificial Intelligence Act+2](#)

中 — 护栏对接硬法：在职场/校园场景禁止滥用情绪识别，并遵守欧盟《AI 法案》与**欧洲委员会《AI 框架公约》**的权利优先义务。 [EUR-Lex+1](#)

6.4.3 Interfaith Compatibility Framework

6.4.3 全球多信仰兼容架构

EN — To coexist with diverse faiths, BrainCoin adopts a **four-layer** framework:

- **Soul Layer:** *Theology remains sovereign*. BrainCoin makes **no metaphysical claims**; it measures biological activity only.
- **Mind Layer:** Consciousness is treated as *dignified*, not commodified; participation is voluntary and opt-out.
- **Social Layer:** Rewards align with *ethical conduct* (e.g., public-goods burns), not belief.
- **Global Layer:** An **Interfaith Advisory Council (IAC)** engages recognized dialogue platforms (e.g., **UNAOC**) for continuous consultation. [unaoc.org](#)

中 — 为与多元信仰共存，脑币采用四层架构：

- 灵魂层：神学自治。脑币不提出形而上判断；仅量化生理信号。
- 意识层：将意识视为具尊严而非商品；参与自愿且可退出。
- 社会层：奖励对齐伦理行为（如公共品销毁），与信仰立场无关。
- 全球层：设立宗教咨询理事会（IAC），并与现有对话平台（如 UNAOC）常态互动。 [unaoc.org](#)

EN — This framework upholds **ICCPR Article 18**—freedom of thought, conscience and religion—by ensuring neutrality toward beliefs and protecting non-participants from disadvantage. [OHCHR](#)

中 — 该框架以**《公民权利和政治权利国际公约》第 18 条**为准绳，通过信仰中立与不参与不受损保护来落实思想、良心与宗教自由。 [OHCHR](#)

6.4.4 Coordination with Religious Law

6.4.4 宗教法与意识权的协调

EN — When integrating into national systems that recognize **Canon Law, Sharia, Dharma** or comparable bodies, BrainCoin proposes:

1. **Interfaith Advisory Council (IAC)**: recognized faith leaders advise on boundaries (opt-in, pastoral concerns, youth protections).
2. **Freedom of Mind Clause**: all statutes/treaties include a right to *decline participation* without economic penalty.
3. **Spiritual Subchains**: optional, **non-transferable** community ledgers for internal ethics/education—implemented with **Soulbound-style** credentials to prevent speculation. [SSRN+1](#)

中 — 当纳入承认教会法、伊斯兰教法、达摩法等体系的国家时，脑币建议：

- 1) 宗教咨询理事会（IAC）：邀请主流宗教代表就边界提出建议（自愿、牧灵关怀、青少年保护）。
- 2) 思维自由条款：各类法律/条约明确保障拒绝参与而不受经济惩罚。
- 3) 灵性副链：面向社群伦理/教育的可选不可转让账本——采用**灵魂绑定（SBT）**式凭证，杜绝投机。 [SSRN+1](#)

EN — Compliance remains rights-centric: **AI governance** under the EU AI Act and **human-rights safeguards** under **CETS 225** provide interoperable baselines for any national concordat. [EUR-Lex+1](#)

中 — 合规坚持权利为先：欧盟《AI 法案》的 **AI 治理**与 **CETS 225** 的人权保障为各国协定提供可互操作的基线。 [EUR-Lex+1](#)

6.4.5 Philosophical Summary

6.4.5 哲学总结

EN — BrainCoin is **not** a substitute for religion; it is a **civilizational protocol** for universal ethics. Religion sanctifies the **soul**; BrainCoin sanctifies **thought**. Together they form a continuum—from spiritual transcendence to cognitive equality—under a shared guarantee of **freedom of thought and dignity**. [OHCHR+1](#)

中 — 脑币不是宗教的替代，而是面向普遍伦理的文明底层协议。宗教赋予灵魂以神圣；脑币赋予思想以尊严。二者共同构成从精神超越到认知平权的连续体，并受思想自由与人类尊严的共同保障。 [OHCHR+1](#)

Implementation Notes

落地说明

EN — The protocol commits to **belief neutrality, opt-in only, no content decoding, no emotion recognition in schools/workplaces, and local-only raw brain data with ZK proofs**—reflecting UNESCO/OECD guidance and binding European law. [Artificial Intelligence Act+3OHCHR+3OECD AI+3](#)

中 — 协议层承诺：信仰中立、自愿参与、不解码思想内容、学校/职场禁用情绪识别、原始脑数据仅本地保存并以零知识证明链上公示——呼应 UNESCO/OECD 指引与欧洲现行法。

6.5 Societal Transformation & Civilizational Transition

6.5 社会结构与文明转型

6.5.1 From Material Economy to Conscious Civilization

6.5.1 从物质经济到意识文明的跃迁

EN — BrainCoin reframes prosperity from **possession** to **cognition**. Classical industrial orders were organized around energy and capital; the knowledge era elevated data and platforms; a consciousness-based economy centers **measurable, rights-preserving thought** as the generative force. This progression echoes long-standing analyses of the knowledge economy and collective intelligence: when institutions and tools reward how we **think together**, societies move from hierarchy to **co-creation**. [PubMed+3OECD ONE MP+3OECD ONE MP+3](#)

中 — 脑币将繁荣的基点由占有转向认知。工业时代以能源与资本为纲；信息时代抬升数据与平台；意识经济则把可计量且权利友好的思考作为生成性力量。这一演进与知识经济及群体智能研究相呼应：当制度与工具奖励我们如何共同思考时，社会便从等级走向共创。 [PubMed+3OECD ONE MP+3OECD ONE MP+3](#)

EN — In this paradigm: **value** arises from the **quality of attention and problem-solving**, not asset endowment; **labor** is cognitive participation bounded by health (BRC caps); and **ecology** becomes a first-class constraint via resource-anchored accounting (SEEA). Humanity evolves from *Homo Economicus* to *Homo Cogitans*. [World Health Organization+1](#)

中 — 在此范式中：价值源自注意力与解题能力而非资产禀赋；劳动是受健康上限（BRC）约束的认知参与；生态通过资源锚定核算（SEEA）成为一等约束。人类由 *Homo Economicus* 进化至 *Homo Cogitans*。 [World Health Organization+1](#)

6.5.2 Restructuring of Social Institutions

6.5.2 社会制度的再结构化

EN — BrainCoin drives a structural rewrite across core institutions:

- **Economy:** from capital accumulation to **cognition-driven creation**; prices align with ecological accounts (SEEA EA).
- **Politics:** from representation to **neuro-democracy**—citizens co-fund public goods via **BGBS** (burn-audited budgets).
- **Education:** from information transfer to **brain-based learning** (activation, reflection, creativity).
- **Labor:** from time-for-pay to **workload-for-reward** (PoNP/PoBW with BRC).
- **Welfare:** from redistribution to **Universal Cognitive Income (UCI)**—self-generated, non-transferable rewards for authentic thought. seea.un.org

中 — 脑币引发关键制度的重写：

- **经济：** 由资本积累转向认知生成；价格对齐生态账本（SEEA EA）。
- **政治：** 由代议制转向神经民主——公民通过 **BGBS** 共投公共品（销毁审计的预算）。
- **教育：** 由知识灌输转向脑基础学习（激活、反思、创造）。
- **劳动：** 由“以时换薪”转向“以脑换币”（PoNP/PoBW + BRC）。
- **保障：** 由再分配转向普遍脑币收入（UCI）——对真实思维的自生成、不可转让回报。 seea.un.org

EN — UCI is **not** a fiscal stipend; it is a protocol outcome: minting from PoNP/PoBW within health limits, audited on-chain via ZK-Receipts and adjusted by **Neural Difficulty**

Adjustment to sustain macro-balance. [CIS UPenn](https://cis.upenn.edu)

中 — UCI 不是财政给付，而是协议产出：在健康上限内，按 PoNP/PoBW 产币，以零知识收据链上审计，并由神经难度调节维持宏观平衡。 [CIS UPenn](https://cis.upenn.edu)

6.5.3 Social Stratification & Cognitive Equality

6.5.3 社会分层与认知平权

EN — Because BrainCoins are **non-transferable and non-inheritable** (SBT-style credentials for services/entitlements), wealth concentration by hoarding or legacy weakens. Opportunity links to **current cognitive engagement**, not lineage. To prevent new inequities (device access, health variance), the protocol funds **access subsidies** and publishes an **Equity**

Dashboard (median output by region/gender/disability/income) to monitor fairness. [Metaneo](https://metaneo.org)

中 — 由于脑币不可转让、不可继承（以 SBT 式凭证提供服务/资格），通过囤积或继承形成的财富集中显著减弱。机会关联于当下的认知投入而非出身。为避免新不平等（设

备可得、健康差异），协议设立接入补贴并发布公平仪表盘（按地区/性别/残障/收入的产出中位数）以监测纠偏。[Metaneo](#)

EN — Cognitive equality also requires **rest and safety**. BRC caps respond to evidence that **long working hours elevate cardiovascular risk**; therefore, reward schedules feature diminishing returns and mandatory off-cycles. [World Health Organization+2ScienceDirect+2](#)
中 — 认知平权同样依赖休息与安全。BRC 上限对应证据：长工时增加心血管风险；因此奖励曲线采取递减，并配置强制离线周期。[World Health Organization+1](#)

6.5.4 Societal Ethics & Civilizational Mission

6.5.4 社会伦理与文明使命

EN — The age of consciousness needs a compact of three universal principles: **Freedom of Thought** (no coercion, opt-in), **Respect for Cognitive Work** (recognize authentic effort beyond market prices), and **Eco-Cognitive Balance** (align every mint/burn with ecosystem accounts). These are **principles of coexistence**, not instruments of control. [ResearchGate+2UNESCO+2](#)
中 — 意识时代需要三条普遍原则：思想自由（自愿、无强迫）、尊重认知劳动（承认超越市场价格的真实投入）、生态—认知平衡（每一次产币/销毁与生态账本对表）。它们是共存的准则，而非控制的手段。[ResearchGate+2UNESCO+2](#)

6.5.5 The Three-Phase Civilizational Transition Model

6.5.5 文明转型的三阶段模型

EN — **Phase I — Emergence (2025–2035)**: experimental BrainCoin pilots; community-scale mining; AI-assisted authenticity; SEEA-aware pricing trials.

Phase II — Integration (2035–2050): fusion of global resource ledgers and cognitive rewards; neuro-democratic budgeting at municipal/national levels.

Phase III — Conscious Civilization (2050+): a Global Cognitive Mesh with **reliability-weighted** consensus and widespread **UCI**, under rights-first governance. [seea.un.org](#)

中 — 阶段一—萌芽期（2025–2035）：脑币试点；社区级挖矿；AI 真伪辅助；SEEA 定价试验。

阶段二—形成期（2035–2050）：全球资源账本与认知激励融合；市/国两级的神经民主预算。

阶段三—意识文明（2050+）：构建可靠度加权的全球脑网与广义 **UCI**，以权利优先的治理为底座。[seea.un.org](#)

EN — Throughout all phases, **collective intelligence** is the strategic asset: diverse, socially sensitive teams systematically outperform high-IQ individuals, which validates BrainCoin's emphasis on **co-creation** and collaboration rewards. [PubMed](#)
中 — 在各阶段中，**群体智能**是关键资产：多元且具社会感知力的团队系统性优于高智商个体，这印证了脑币对**共创与协作奖励**的强调。 [PubMed](#)

6.5.6 Final Vision: The Earth Charter of Consciousness

6.5.6 最终愿景：意识的地球宪章

EN — When the protocol matures, human identity is not sorted by nation, class, or creed, but by **contribution of consciousness**. Each person becomes a node in Earth's collective mind; each thought a pulse in the planet's evolving intelligence. BrainCoin is not the end of money—it is the **beginning of conscious civilization**, where **mint** ↔ **burn** cycles keep value aligned with ecology and dignity. [seea.un.org](#)
中 — 当协议成熟之时，人类的身份不再由国家、阶级或教派界定，而由**意识贡献**来界定。每个人都是地球集体心智的节点；每一次思考都是文明演化的脉冲。脑币不是货币的终点，而是**意识文明的起点**；**产币↔销毁**的循环，使价值始终与生态与尊严同频。
[seea.un.org](#)

6.6 Compliance Toolkit

6.6 合规工具箱

EN — **Privacy engineering**. (a) **Local-only raw EEG**; (b) **FL** with secure aggregation; (c) **DP** releases with ϵ -tracking; (d) **ZK-Receipts** for mint/burn; (e) **Article 9** DPIA templates; (f) **AI Act** risk mapping (no workplace/school emotion recognition). [arXiv+2Computer and Information Science+2](#)
中 — **隐私工程**。(a) 原始 **EEG** 仅端侧存储；(b) 联邦学习+安全聚合；(c) 差分隐私发布并记录 ϵ ；(d) 产币/销毁用**零知识收据**；(e) **第9条**专用 DPIA 模板；(f) **AI 法案**风险映射（不得在职场/学校实施情绪识别）。 [arXiv+2Computer and Information Science+2](#)

EN — **Governance**. Dual boards: **AI Governance Council** (technical safety) × **Human Ethics Board**(rights/neuro-ethics); annual **transparency reports** and treaty-alignment checks (UNESCO 2021; OECD 2019; CETS 225). [Naavi+2OECD AI+2](#)
中 — **治理机制**。双委员会：**AI 治理委员会**（技术安全）× **人类伦理委员会**（权利/神经伦理）；发布年度**透明报告**并进行条约一致性审查（UNESCO 2021；OECD 2019；CETS 225）。 [Naavi+2OECD AI+2](#)

6.7 Chapter Summary

6.7 章节总结

EN — From **material accumulation** to **cognitive creation**; from **representation** to **neuro-democracy**; from **redistribution** to **UCI**; from **time-based labor** to **thought-based value**; and from **externalized ecology** to **SEEA-aligned equilibrium**.
中 — 从物质积累到认知创造；从代议政治到神经民主；从再分配到 UCI；从以時計酬到以脑计值；从外部化生态到 SEEA 对表的均衡。

EN — BrainCoin’s legality rests on **rights-first design**: proof-without-exposure (zk-SNARKs), **GDPR Art. 9** compliance for neurodata, neurorights-aware rollouts (Chile/EU), bans adherence under the **EU AI Act**, and treaty alignment with **CETS 225**. The system aspires not to replace legal tender but to **coexist** as a rights-preserving, privacy-first economy of thought. [Portal+4UCL Discovery+4GDPR+4](#)

中 — 脑币的合法性基于权利优先的系统设计：零知识的无暴露证明、对 **GDPR 第 9 条** 的神经数据合规、以神经权利为前提的试点（智利/欧盟）、遵守欧盟《**AI 法案**》禁令，并与 **CETS 225** 保持条约一致。它并非取代法币，而是以权利与隐私优先的方式，与现行体系并行共存。 [Portal+4UCL Discovery+4GDPR+4](#)

Selected academic/official sources | 精选学术/权威来源

- **ICCPR Art. 18** (freedom of thought). [OHCHR](#)
- **EU AI Act – Article 5 prohibitions** (incl. workplace/school emotion recognition). [Artificial Intelligence Act+2European Parliament+2](#)
- **Council of Europe Framework Convention on AI (CETS 225, 2024)**. [Portal+1](#)
- **UNESCO (2021) Recommendation on AI Ethics**. [Naavi+1](#)
- **OECD AI Principles (2019)**. [OECD AI+1](#)
- **Chile neurorights (Law 21.383, 2021)**. [Carey+1](#)
- **Differential Privacy (Dwork & Roth)**. [Computer and Information Science](#)
- **Federated Learning (McMahan et al.)**. [arXiv](#)
- **zk-SNARKs (Groth’16)**. [UCL Discovery](#)
- **Neuroethics priorities (Yuste et al., Nature 2017)**.

Chapter 7 | Manifesto for the Age of Conscious Civilization (Revised)

第七章 | 未来宣言：意识文明的诞生（修订版）

7.1 Prologue: The End of Money, The Beginning of Consciousness

7.1 序言：金钱的终结，意识的开始

EN — Since the dawn of civilization, money has been both instrument and restraint: it enabled exchange, yet narrowed intention. BrainCoin reframes wealth from **possession** to **cognition**, aligning value with verified effort (PoBW/PoNP) and privacy-preserving proofs rather than accumulation. It is not the end of all money, but the **awakening of value** to its mental source.

中 — 自文明伊始，货币既是工具亦是枷锁：它促成交换，却束缚意图。脑币把财富从占有重心转向认知，以可验证的努力（PoBW/PoNP）与保护隐私的证明取代积累逻辑。它不是一切货币的终结，而是价值回归心智源头的觉醒。

7.2 The Mission: To Measure Every Thought That Matters

7.2 时代使命：让每一次思考都被计量

EN — Labor is shifting from physical throughput to **conscious engagement**. BrainCoin's mission is to ensure that each authentic act of thinking is **recognized, recorded, and rewarded**—valuing **authenticity** over velocity and **depth** over volume, while admitting uncertainty through TMV-Spec and confidence-gated rewards.

中 — 人类劳动正由物理产出转向意识投入。脑币的使命，是让每一次真实思考都被承认、记录与回馈——以真实性优先于速度、以深度优先于数量，并通过 TMV-Spec 与置信门控在不确定性中稳健计量。

7.3 The Technical Manifesto: Thought as the New Energy

7.3 技术宣言：让思想成为能量的源泉

EN — The protocol tracks **cognition, not content**: EEG-based intensity (PoNP) and workload (PoBW) mint tokens under health-capped schedules (BRC). The ledger records **zero-knowledge receipts** instead of raw signals. Our horizon is a **Conscious Energy Grid**, where each person's thinking contributes safely and privately to the collective intelligence of Earth.

中 — 协议计量认知而非内容：以 EEG 强度（PoNP）与工作量（PoBW）在健康上限（BRC）下产币。账本仅记录零知识收据而非原始信号。愿景是构建一张意识能网，让每个人的思考在安全与隐私中汇入地球的集体智能。

7.4 The Social Manifesto: Equality Begins in the Mind

7.4 社会宣言：让平等回归思想本身

EN — BrainCoin proposes **cognitive equality**: value emerges from **what you contribute mentally**, not what you inherit materially. It is neither capitalism nor socialism but **cognitivism**—a rights-preserving, non-transferable economy that rewards creation, recognizes care and learning, and publishes inclusion metrics to monitor fairness.

中 — 脑币提出认知平等：价值源自你的心智贡献，而非物质继承。它既非资本主义也非社会主义，而是认知主义——在权利保障与不可转让框架下奖励创造，承认照护与学习，并以公开的包容性指标守望公平。

7.5 The Ecological Manifesto: The Planet Thinks With Us

7.5 环境宣言：让地球与思维共呼吸

EN — Prices are **resource-anchored** (SEEA/LCA): every token minted and burned is reconciled with ecological accounts, so consciousness and sustainability converge. Scarcity is allocated by **burn auctions** and public-goods are delivered via **Service SBTs**, ensuring access without speculative trade.

中 — 价格以资源为锚（SEEA/LCA）：每一份产币与销毁均与生态核算对表，使意识与可持续汇合。稀缺以销毁竞拍分配，公共物品以服务 SBT 提供，保证使用权而非可交易性。

7.6 The Civilizational Manifesto: Humanity as Its Own Creator

7.6 文明宣言：让人类成为自己的创造者

EN — When thought becomes the unit of value, law must safeguard **freedom of thought**, and technology must default to **privacy-by-design**. AI joins not as a sovereign owner but as **collaborator-verifier-balancer**, bound to human guardianship and **forbidden to accumulate** wealth—so the economy becomes a grammar of awakening, not domination.

中 — 当思维成为价值单位，法律须护卫思想自由，技术须以隐私即设计为默认。AI 不是

主权所有者，而是协作者—验证者—平衡者，受人类监护并不得积累财富——让经济成为觉醒的语法而非支配的工具。

7.7 The Earth Pledge of Conscious Civilization

7.7 意识文明的地球誓言

EN — We pledge: **no coercion of thought** (voluntary participation and opt-out), **no surveillance of minds** (local processing, DP/FL, ZK-Receipts), and **no ecological externalization** (SEEA-anchored accounting). When every being can think freely and live creatively, the ledger becomes a mirror—Earth, a living mind.

中 — 我们誓言：不强迫思想（自愿参与与可退出）、不监控心智（本地处理、DP/FL、ZK 收据）、不外部化生态成本（SEEA 锚定核算）。当众生自由思考、自由创造之时，账本成为镜面——地球成为有思的生命。

7.8 Epilogue: To the Future Humanity

7.8 结语：致未来的人类

EN — BrainCoin is a beginning, not an end: a parallel, rights-preserving economy that **measures meaning before might, perception before production**. For the first time in history, **wealth keeps a conscience**—and civilization keeps its soul.

中 — 脑币是起点而非终点：一套并行且以权利为先的经济体系，先量意义，后量力；先见知，后产出。人类历史上第一次，财富拥有良知——文明保有灵魂。

Selected sources (academic / official)

- **UN SEEA** (System of Environmental-Economic Accounting) — resource-anchored pricing and ecosystem accounts.
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- **OECD AI Principles (2019)** — inclusive growth, human-centered values.
- **Dwork & Roth (2014)** *The Algorithmic Foundations of Differential Privacy*.
- **Nunez & Srinivasan (2021)** *Electric Fields of the Brain*; **Cohen (2014)** *Analyzing Neural Time Series Data*.
- **Ienca & Andorno (2017)** neurorights discussion, *Nature Human Behaviour*.